The Sorting Guide
Kildesorteringsguiden

Bachelor Project

Brage Steen & Regina von Koch
Oslo Metropolitan University
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Abstract

The Sorting Guide project sets out to increase the rate of recycling. In the complex system of systems that is recycling, there is a major responsibility resting on the household's shoulders – the first line of recycling. Most households are uneducated on the topic of recycling and not equipped with the resources necessary to recycle correctly. Through the empowerment of this group, the recycling system's most obvious leakage can be reduced and the rate of recycling increased.

By using qualitative methods, primarily observation and interviews, an extensive body of data has been collected to investigate the issues surrounding recycling at home. In addition, design probes and desktop research has been included to ensure a level of triangularity. This has provided the project with a high rate of validity. The collected data has been thoroughly mapped out and key findings identified. These were lack of availability, lack of education and lack of trust.

In cooperation with IKEA, the project's main partner, a service platform and Gigamap has been developed. The final service arose from a comprehensive ideation process where many ideas and concepts where created and discarded. The platform serves as the database for local recycling guidelines and IKEA products. From there, the service provides the user with personalized information on how to sort correctly and what recycling products are recommended.

Abstract

IKEA's customer base represents a large share of Norway's population which provides IKEA with a unique opportunity and significant responsibility to affect positive change in society. Consequently, the potential positive impacts from the proposed Sorting Guide is significant. While many products, services and other solutions attempt to reduce waste and make recycling easier – the Sorting Guide provides a platform for all key stakeholders in private households by empowering them to sort correctly. Through a higher rate of recycling, the need for virgin products is reduced and the level of re-use increases, leaving a positive effect on both society, business, and environment.

Preface

This is the ending bachelor thesis for the three year Product Design education at Oslo Metropolitan University. The project has been a great learning experience and we highly appreciate the help of advisors, fellow students and professors.

We would like to extend our gratitude to Johannes Daae, our contact at Grønt Punkt and Kenneth Skauge, our contact at IKEA, for the insightful collaboration and feedback. Additionally, we would like to thank Anna Kirah, for great help along the way.

Regina von Koch

Regina von Koch



Brage Steen

May 20th 2020
Oslo, Norway

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Chapter 1

Introduction

- Background
- Importance of the Project
- The Team
- Partners & Involved
- Design Question
- Design Process



Background

The increase in population and consumption has it demands. We extract resources from the earth at a faster rate than it can re-generate. This is illustrated by Earth Overshoot Day - the day in the calendar year when the world has used up all the regenerative resources of that year. In 2019, it was on July 29th! (Earth Overshoot Day, 2020).

> *"It's a global emergency"*
>
> - Pope Francis
> Earth Overshoot Day (The Paris Agreement, 2020)

The Paris Agreement is a contract signed by 186 countries to combat climate change (The Paris Agreement, 2020). If we set out to reach the goals set by this agreement, a key element would be to change the perspective in which waste is viewed and handled. Waste is not waste, but a used resource that has a capacity of re-use and renewal. "This can be achieved through long-lasting design, maintenance, repair, reuse, remanufacturing, refurbishing and recycling" (Geissdoerfer, 2017, p.6).

In Norway, 21% of the waste is from households. Today over 60% of the household waste is not sent to recycle after the waste management facility (SSB, 2020). A large part of this is due to errors made by the first line of recycling - the people at home - who makes sorting decisions and handles waste first-hand. When sorting and handling errors are made at this level, it creates major challenges for the recycling process.

Project Focus

The project focus is to empower the first line of recycling in order to increase the rate of recycling and reduce the amount of general waste.

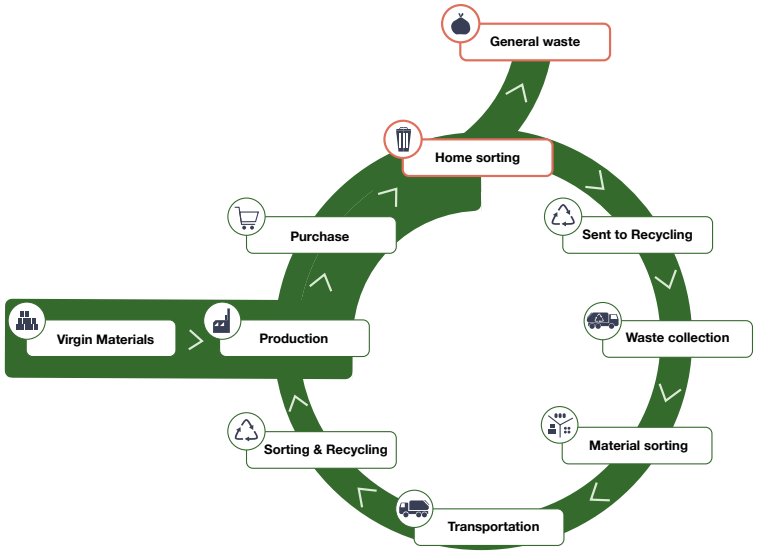


Figure 1: Leakage of Plastic. , Illustration inspired by Grønn Punkt Regina von Koch, Lillestrøm. 2020



Figure 2: Haraldrud Sorting Station (Photo: Brage Steen)

Sustainable Development Goals

Importance of the Project

The United Nations members have developed seventeen Sustainable Development Goals (SDGs) for 2030 that focuses on protecting the planet, ending poverty, and improving people lives around the world (United Nations, 2020).

This project aims to partake in UNs effort and have chosen its' own SDGs to focus on.

#11 Sustainable cities and communities

Today, over 2 billion people do not have access to waste collection services (United Nations, 2020).

In Norway, these services are available, but they still struggle to meet the recycling goals. In this project we will investigate how to contribute to solve this problem and were we do most change by using our skills as designers.

#13 Climate Action

The UN to reach have a goal of 45% reduction of global carbon emissions in the period 2010 - 2030 (United Nations, 2020).

This project aims to deliver a higher rate of recycling which may contribute to less carbon emissions. It is more efficient to recycle materials than produce new once.



Figure 4: SDG 11



Figure 5: SDG 8

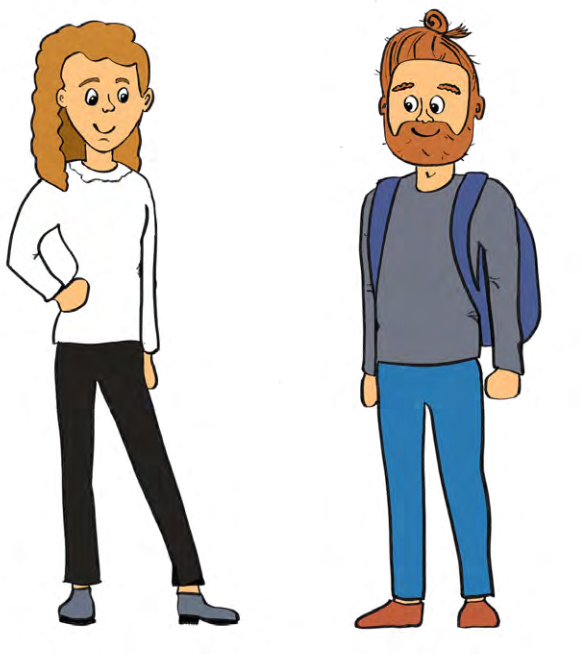


Figure 3: SDGs

Meet the Team

The team consists of two members, Regina von Koch and Brage Steen, whose motivation for working together comes from a shared interest of recycling, sustainability and making a difference.

Regina



Brage

Figure 6: Illustration of the team



Figure 7: Team Photo (Photo: Øivind Lundestad)

Partners

IKEA

IKEA is the main partner for this project. IKEA is one of the world's largest providers of furniture, kitchen, and home accessories. Due to their large customer base who all have recycling solutions in their home, they are in a great position to affect change. The project is based on the partnership with IKEA Norway and the delivered service is developed with them in mind. Our contact person is Kenneth Skauge, Sustainability Business Partner.

Grønn Punkt

Grønt Punkt has served in an advisory role in the project. They are a recycling facilitator in Norway, and their involvement has been vital to see our problem area in a larger and more advanced context. Our contact person is Johannes Daae, Resposibal EcoDesign.

Others

In addition to the project's main partners, we also had other external resources involved.

Halogen Design
Anna Kirah, Design Anthropology and Psychology Specialist

OBOS
Sigbjørn Windingstad, Senior Industrial Designer at Inventas

Loop
Hildegunn Bull Iversen, Managing Director

UX Designer
Runar Normark

GIGA-Map

Waste in the World

For a better understanding of the current waste situation, we have developed a research map explaining the world of waste. This has provided a foundation for choices made throughout the project.

To see the research map and learn more about the world of waste; click the button below.

Click Here



THE **PARIS** AGREEMENT

The Paris Agreement is a contract signed by 186 countries around the world where the goal is to keep the temperature level under 2 degrees from pre-industrial times. The main focus is to reduce greenhouse gasses and emotion. If we set out to reach the goals set by this agreement, the world needs to move to a more sustainable waste management system. A key element in this change is the perspective in which waste is viewed. Waste is not waste, but a used resource that has a capacity of re-use and renewal. (4)

"This can be achieved through long-lasting design, maintenance, repair, reuse, remanufacturing, ***refurbishing and recycling****"* (22)

0%
EMISSION GOAL





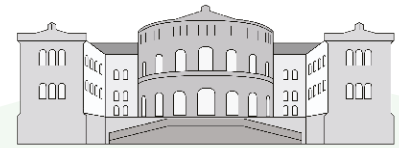


"Reducing loss of resources from waste management is to strengthening the circula economy in Europe" (24)

Recycling by the Nu

EUROPE
BY 2035
65%

NORWAY
TODAY
37%



#2

Norway has the second most waste per capita in Europe

739 kg
Total waste per capita in Norway (6)



Figure 8: Giga-map

GOVERNEMENT
REGULATIONS

Business resonsibility

From January 2018, Businesses that produces more than 1000 kg of a certain packaging has to be member of a valid recycling company that is verified by Miljødirektoratet. (8)

Grønt punkt and Norsirk

The packaging producers have to pay a fee and be members of Norsirk or Grønn Punkt. They take care for that the packaging the businesses add to the market is being delivered to recycling and is recycled. For example, Grønn Punkt reports how much waste is delivered to Germany and Germany report back how much is actually recycled. (9)

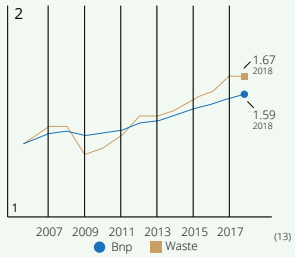
Municipality responsibility

Forurensningsloven states that all the municipalities in Norway have the responsibility for collecting waste generated by the household. In 2017, 39% of all household was sent to recycling. (10)

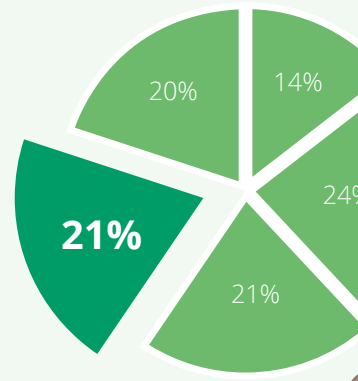
LAGGING
BEHIND

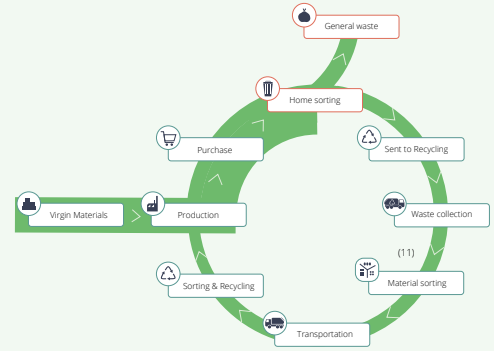
According to Norges miljømål 4.3, the growth of amount of waste should be a lot lower than the economic growth in the country, however, the Norway's gross national product increased with 2% from 2016 to 2017 while the growth of the amount of waste increased with 3%. These numbers shows that Norges miljømål 4.3 has not been reached. (s40) (7)

Devolopment in
waste compared to GDP.



(13)





(11)

LEAKAGE IN THE SYSTEM

Today over 60% of the household waste is not sent to recycle after the waste management facility. A large part of this is due to errors made by the first line of recycling - the people at home - who makes

Design Question

During the process we have had a variety of design questions. To understand the design process, it is important to know how the design question has been re-defined over time. The different design questions will be presented throughout the report.

Our final design question is:

?? *"How can we improve household recycling by guiding people looking for recycling solutions, to information and offerings based on their household needs, that leads to minimizing general waste?"* ?

Design Process

The design process is based on the four core activities of service design as illustrated in the figure to the right. Research, ideation, prototyping, and implementation processes continuously overlap and paves way for a creative, yet structured project (Hormess, Lawrence, Schneider & Stickdorn, 2011, p.185).

While this model has been exercised as the framework of this report, the processes are iterative and has enabled a divergent and convergent way of working - where we started with a divergent research session of gaining valuable insights through the use of a variety of methods and then a convergent phase trying to map and connect the most valuable insights.

By continuously moving back and forth between the spaces, and applying the ideas of divergent and convergent thinking, we were able to refine the ideas and further investigate different directions.

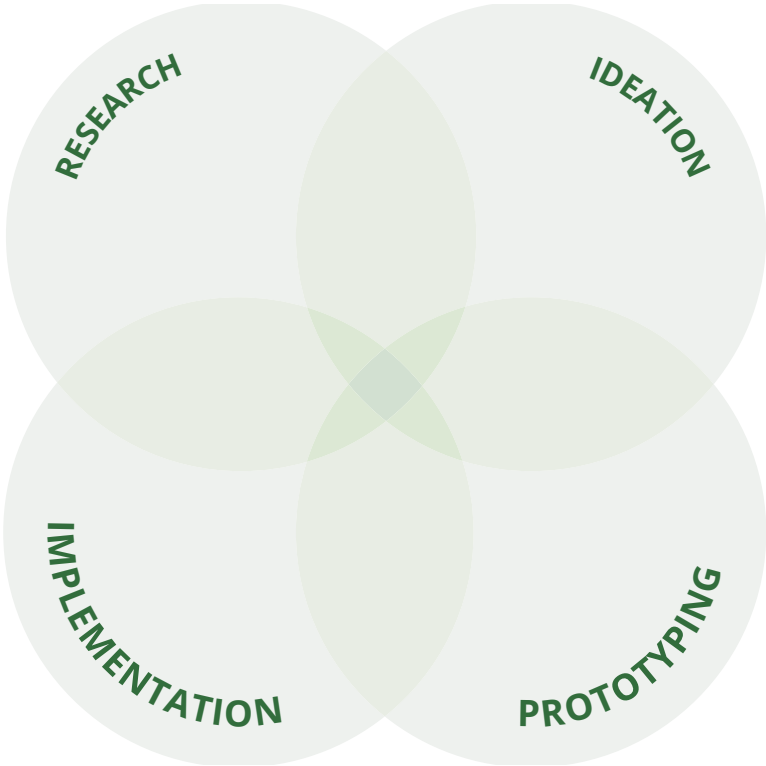


Figure 9: The four core ativities of Service Design. Illustration inspired by This is Service Design Doing (Hormess, Lawrence, Schneider & Stickdorn, 2011, p. 185)

Chapter 2

Research

- Overview
- Gigamap
- Field Trips
- Home Visits
- Workshop



Figure 10: Workshop at Sintef

Overview of Research

The research process has been conducted using a qualitative approach. Qualitative methods provide a good starting point for gaining empathy and finding insights (Brown, 2009).

This method of research allows for a more dynamic, interactive data collection process where it is important to research the topic from different angles and is more likely to catch the complexity of the problem (Kirah, 2020).



Figure 11: Timer on cameras are difficult. (Photo: Own)

Desktop Research & Gigamapping

Desktop research and Gigamapping have been key methods in the starting phase to define the direction of the project. It was important to be critical of the information we gathered through research and find sources that viewed issues from opposing perspectives.

Findings

- People have strong opinions about the recycling issue
- The complexity of the recycling system is vast and can be described as a, ”system of systems”.
- Many local apadations to the recycling systems
- The ”first-line of recycling” is the most vulnerable point in the system, yet also maybe the most important.



Figure 12: Gigamapping (Photo: Own)

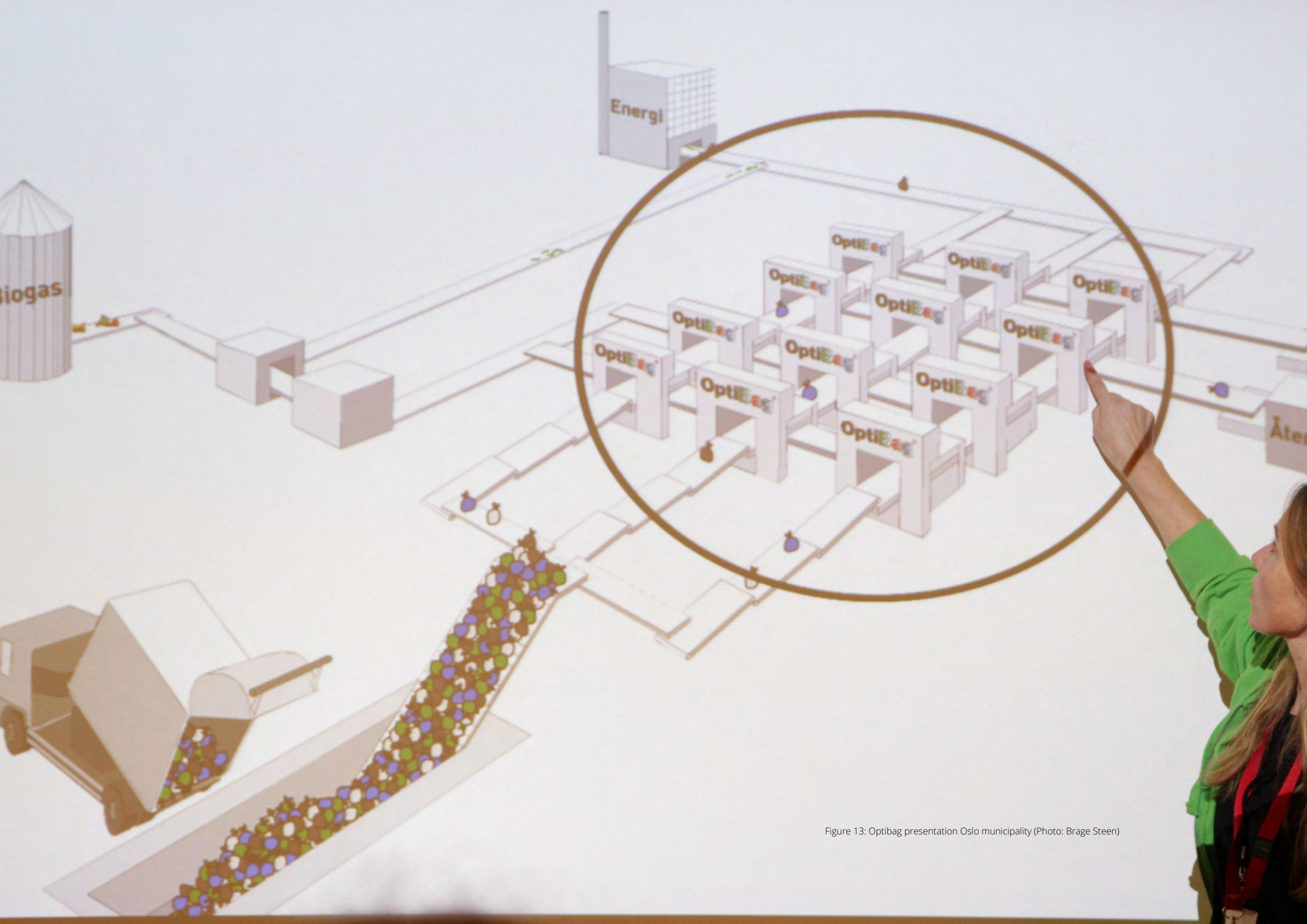


Figure 13: Optibag presentation Oslo municipality (Photo: Brage Steen)

Recycling Facilities

Field trip

To get a better understanding of the recycling system and how waste is handled in Norway, we went to visit the RoAF recycling center and the main recycling facility in Oslo. The RoAF automated equipment is state-of-–the-art machinery and the facility is among the most technologically advanced in Europe.

Findings

- The blue plastic bags in Oslo is automatically sorted to general waste if they weigh more than 600 grams.

- There is a big difference in the efficiency of recycling systems between municipalities. The average amount of recycled plastic per person per year is:

RoAF: 18 of 24 kg
Oslo: 3,9 of 24 kg

- The RoAF system removes some of the home sorting decisions as their recycling facility can do this on their behalf - this puts less pressure on people at home.



Figure 14: RoAF Recycling Center (Photo: Regina von Koch)



Figure 15: Oslo Recycling Center (Photo: Brage Steen)

Complex system

Recycling in Norway work differently based on which municipality you live in. We have chosen Oslo as main focus in this project. Oslo has many inhabitants and is one of the worst municipalities in Norway to recycling certain waste types such as plastic. (Kantar, 2019)

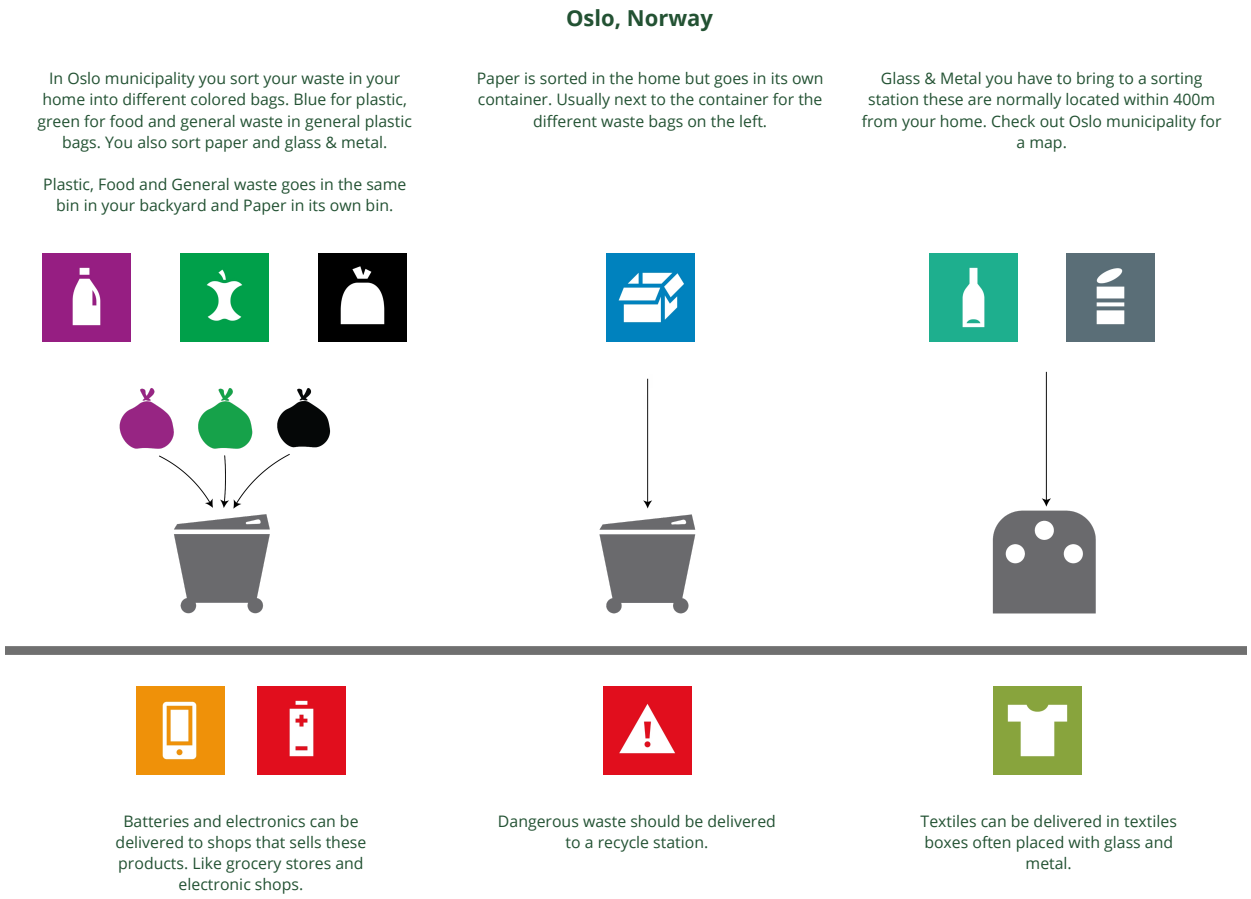


Figure 16: Overview of Oslo system



Figure 17: Researching kitchens (Photo: Regina von Koch)



Figure 18: Recycling solutions (Photo: Brage Steen)



Figure 19: Smart solution (Photo: Brage Steen)

Market Research

To understand how people easier can recycle at home, it was essential to do field research on existing products. We went to look at kitchens and recycling solutions from six different high-end and low-end kitchen suppliers.

Findings

- Majority of systems are located under the sink
- No solution for e-waste (batteries, light bulbs, etc.)
- No recycling options for bathrooms or other parts of the home
- Advanced bins included air filter for food waste to minimize smell



Figure 21: Observation IKEA (Photo: Regina von Koch)

IKEA

We visited IKEA at Furuset, Slependen, and the newly opened planning studio in central Oslo. The kitchen sales process was observed at a distance and we also went through the kitchen ordering process several times. When buying a new kitchen we wanted to see how recycling was included in the process.

> *"We watch what people do (and do not to) and listen to what they say (and do not say)"*
>
> - Tim Brown (Brown, 2009, p.43)

By practicing Tim Brown's approach, we were able to go out of our comfort zone and gain valuable insights (Brown, 2009).

Findings

Customer Focus

Recycling is not a focus for the customer. Most of the time they do not ask about recycling options when ordering kitchens.

Business Priority

Recycling is not a priority for the business. The visual displays for recycling bins where poor and the product options for recycling are presented late, or never, in the kitchen sales process.

Local Differences

Different areas have different waste management systems and therefore have different recycling product needs.

At last, the lid is included for the same price. Our research indicate that most people do not use the lid on the trash bins. Why include them if people don't use them?



Figure 22: Visit to IKEA 2 (Photo: Regina von Koch)



Figure 23: Recycling display IKEA (Photo: Regina von Koch)



Figure 24: Color-marked bins (Photo: Regina von Koch)



Figure 25: Recycling solutions IKEA (Photo: Regina von Koch)

Home visits

People-centered research is an effective way to obtain qualitative data. The team opted to interview and visit fewer people to rather spend more time with each to get into the depth of the matter more thoroughly. When similar answers arose from different interview and observation sessions, the validity of insight was strengthened. Through home visits we got first-hand experience on home recycling process.

Participants

- Couple 45 years old, with 3 children. House
- Couple 40 years old, with 2 children. Apartment
- Male 25 student, living in collective with 6 people. Apartment
- Couple 26, Students. Apartment
- Married 55, one dog, kids moved out. Apartment

> *"The ones who empty the trash are everyday heroes!"*
> - 25-year old student



Figure 26: Home visit (Photo: Regina von Koch)



Figure 27: Food Bag wrong place (Photo: Brage Steen)



Figure 28: Home Sorting (Photo: Brage Steen)



Figure 29: Home Sorting in Oslo (Photo: Brage Steen)

Home visits

The interviews were strategically unstructured so that the subjects would not be familiar with the purpose. It was a priority for the interviewer to ask open-ended questions and allowing the participant to speak freely to not force the conversation in a certain direction.

By not disclosing the purpose of our visits, the data collection would be as close to real-life as possible, rather than the subjects preparing for it by presenting their homes as artificially clean or recycle-friendly.

> *"Actually no point sorting plastic. It gets burned. But we do it anyways"*
>
> Couple 45

Design Probes

To gather more insights from homes, we sent out a set of design probes (Hormess, Lawrence, Schneider & Stickdorn, 2011).

> *"I hate the lid, it´s so annoying"*
> - Student, 25 years old

More pictures and information can be found in Appendix p. 8.



Figure 30: Design probe 1



Figure 31: Design probe 2



Figure 32: Design probe 3



Figure 33: Design probe 4

Mapping of Insights

Home visits and design probes

We drew up a representation of the home visit and printed out pictures from the visits and connected lines to find resemblances and link findings to each other. Another important step was to put up heart post-its showing people´s workarounds - the workarounds are important to understand how and why people create their own solutions to the problem (Kirah, 2020).

Findings

- Availability
- Lack of space
- Uncertainty of recycling
- Difficult to organize system at home

Reflection

It was difficult to differentiate what people felt and said to their actual actions.

More pictures and information of home visits can be found in Appendix p. 9-13.



Figure 34: Mapping of Insights (Photo: Regina von Koch)



Figure 35: Meeting with Loop (Photo: Regina von Koch)

Loop

Field trip

Loop, a non-profit foundation, aims to improve the rate of recycling by the average person. They facilitate a sustainability and recycling focus in schools and public services in Norway. Currently, Loop is developing new recycling icons to provide the user with information for sorting. Their strategy is to implement corresponding icons on product packaging and waste bins to simplify the sorting process.

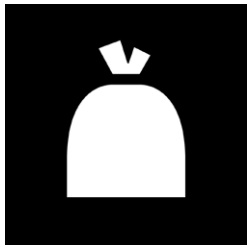
Reflection

What if IKEA chould have stickers with the new colors and icons for better understanding of waste management?



Figure 36: User Testing of the new Recycling Icons (Photo: Hildegunn Iversen)

New Recycling Icons



General waste



Plastic



Food



Paper & Cardboard



Textiles



Metall



Glass



Batteries



EE - Waste



Dangurous waste

Figure 37: New Recycling Icons

Development Goals Forum

Design Sprint

We visited DGF2020 at DOGA where we attended a design sprint hosted by Eggs Design. The purpose of attending was to learn more about Sustainable Development Goals, circular economy and to prepare for upcoming workshops by gathering inspiration.

Reflection

It was fun to see how interesting ideas can form when cross-functional teams are working together. We see the value of including people with different perspectives to gather a wider spectrum of ideas.

DGF
DEVELOPMENT
GOALS FORUM
2020 GLOBALISATION

Figure 38: Development Goals Forum



Figure 39: Design Sprint (Photo: Regina von Koch)



Figure 40: Workshop at Oslo Metropolitan University (Photo: Regina von Koch)



Figure 41: Regina instructing and helping participants (Photo: Brage Steen)

Co-Creation Workshop

We organized two workshops to gather insights (Plumbe et al, 2010, s. 35). The first workshop was for SmartPACK at SINTEF in Oslo, where members from major firms such as Orkla, Norgesgruppen and Bama attended. 27 people participated and the workshop was completed in 40 minutes. The second workshop was at OsloMet University for 30 students and three professors.

Findings

Experts

- Home sorting is time consuming
- Users do not have proper information or knowledge to sort correctly
- There is not enough space in the home for all the recycling bins.
- Users thinks the optimal solution would be to compress the trash.

Students

- Users have a lazy attitude towards recycling and think it is a lot of work
- The collections bins can be located far away (metal and glass)
- Users are experiencing lack of trust to the system
- Users thinks optimal solution would be if there was just one bin for everything.

Co-Creation Workshop

Reflection

Reflection

By using the same method on two different groups, we discovered something interesting. The expert group and student group had a vastly similar response – they both struggled to grasp the issues at hand due to the complexity of the recycling system. Neither group had in-depth understanding of the system, but rather only the matters directly concerned to them.



Figure 42: Workshop OsloMet (Photo: Regina von Koch)



Figure 43: Workshop SINTEF (Photo: Regina von Koch)



Figure 44: OBOS Living LAB (Photo: Regina von Koch)

OBOS Living Lab

Field trip

OBOS, a construction and real estate company, is building an apartment complex in Vollebekk where new solutions and concepts for future living will be tested.

Findings

We gained valuable understanding of trends; people will live in smaller spaces, share resources, and utilize technology more efficiently. Additionally, the team learned that OBOS struggled to find satisfactory recycling solutions for small homes. The waste management system had not been a focus in the Living Lab project.



Figure 45: OBOS Living Lab Presentation (Photo: Regina von Koch)

OBOS Living Lab

Reflection

The OBOS Living Lab concept was innovative and challenged us to be creative. By thinking how people will live in the future we opened our minds to new ways of thinking. It is likely we will have more shared space and co-living areas, which implies more social interaction. This is an opportunity to make recycling a social activity. In addition, we thought about how gamification and advanced technology could impact recycling.

We were inspired and illustrated a possible future scenario, where the following elements were included:

- Waste competition games
- An underground recycling lab
- 3D printing studio where plastic can be recycled locally and turned into products

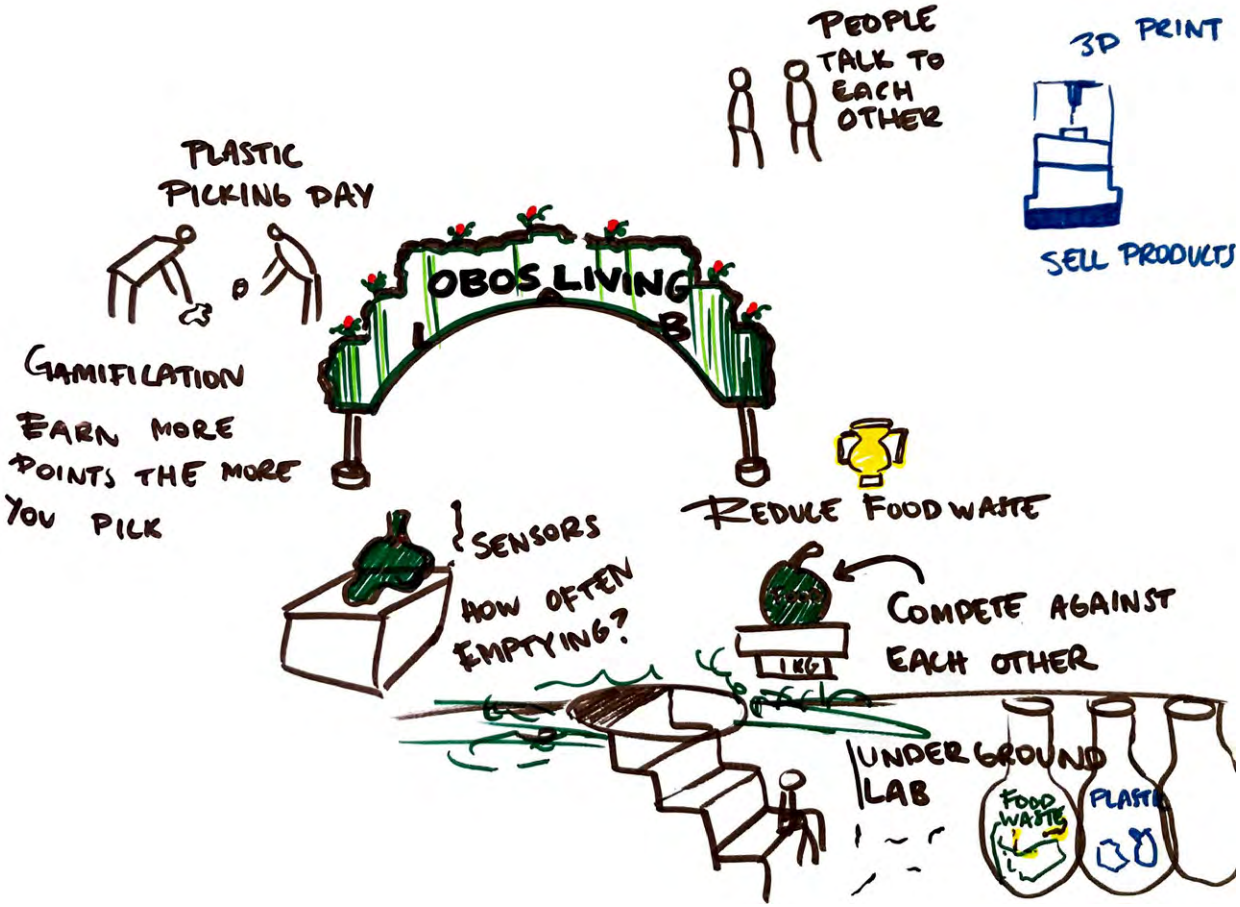


Figure 46: Illustration of ideas



Figure 47: Visiting Grefsenkollen Restaurant (Photo: Regina von Koch)

Commercial kitchen

Field Trip

For the project to get a wider perspective on recycling, we went to visit Grefsenkollen restaurant to observe the recycling process at the business level. This gave us insights from an extreme outlier that can prove to be useful when applied to the general user (Brown 2009. s. 44).

Findings

- Overall, the recycling structure is poor, the placement of the bins is far away from where the waste is generated
- Recycling was not a focus and the employees did not have sufficient knowledge
- Most of the plastic gets sorted in general waste
- Hygiene is important

The restaurants chef was eager for us to come back and help with recycling.

Summary Research Phase

After the primary research phase, we sat down to discuss our collected data and findings. We categorized the findings and evaluated them individually. When certain findings occurred multiple times, it illustrated to us that they were important. We discussed and ranked the most important ones and ended up defining three key findings. These key findings have a high level of validity as they arose from different sources using different methods.

Key Findings



Availability

The availability and placement of the recycling bins determines the recycling percentage in the home. The more space available and the easier placement, the more you recycle. Space determines where in the home people place the recycling bins.



Education

People are uneducated and don't possess sufficient knowledge to recycle efficiently. This is a major barrier to recycling as people don't know where and how to recycle. The complexity of the system and variety of materials confuse and alienate people, distancing them from the issue.



Trust

Personal barriers to recycling include lack of trust to the system. When people don't understand the system, they don't trust or believe in it. Due to the complexity of the recycling system, it is difficult for people to see how they make a difference, which reduces their trust.



Discussing our work and insights with master students working with recycling

Figure 48: Insight Discussion (Photo: Regina von Koch)

Chapter 3

Ideation

- Idea Development
- Idea Selection
- Concept Selection
- Concept Development



Figure 49: Ideation

Overview of Ideation Process

Based on the mapping of insights from the research phase we have been brainstorming ideas. To further explore the problem, the method: asking critical questions, were used (Lerdahl, 2017, p.52). This helped to explore thoughts but also to concretize the critical questions further and was formed into an open design question. It was important for the question to be open because the purpose was to generate ideas.

Design Question

How to optimize recycling in the home for the user?

Key words

Fun, Positive Experience, Easy to use, Space effective

We were able to develop many ideas by using the method Brainwriting pool with external people (Lerdahl, 2017, p. 112)

These ideas were later divided into the following five categories.

- Gamification
- Information
- Simplify life
- Effective Space solutions
- Future



Figure 50: Discussing and developing

Gamification

Ideation

1. Flavored packaging where taste indicates where to sort it

2. Make home sorting a basketball game where the bins lead directly to the outside bins.

3. Rubik's cube waste game to make it fun – but also difficult?

4. Board game about recycling to bring friends or family together

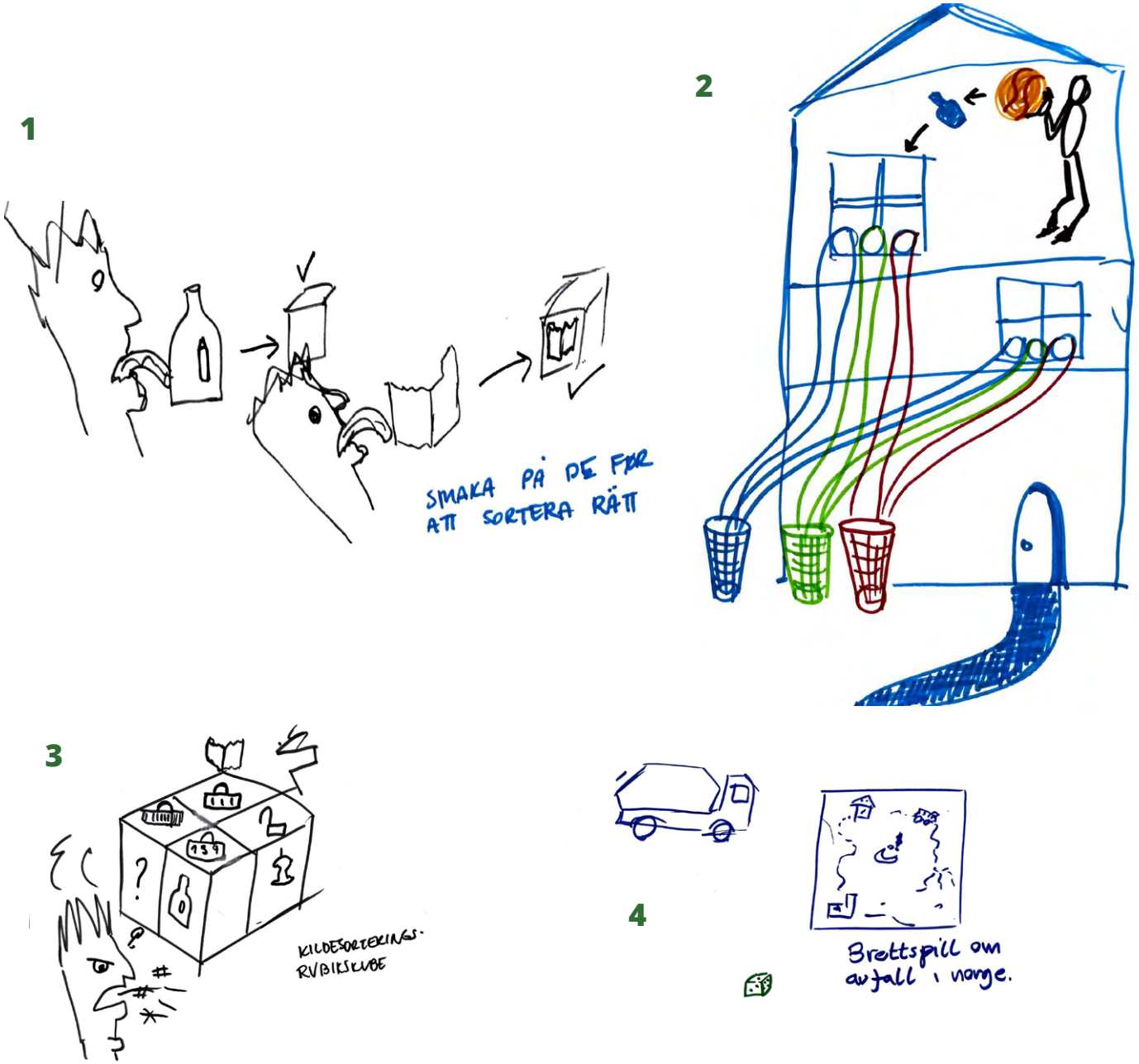


Figure 51: Gamification sketches

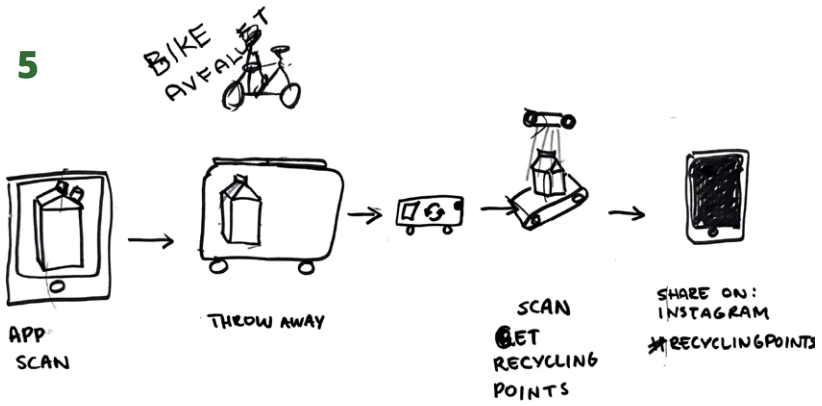
Gamification

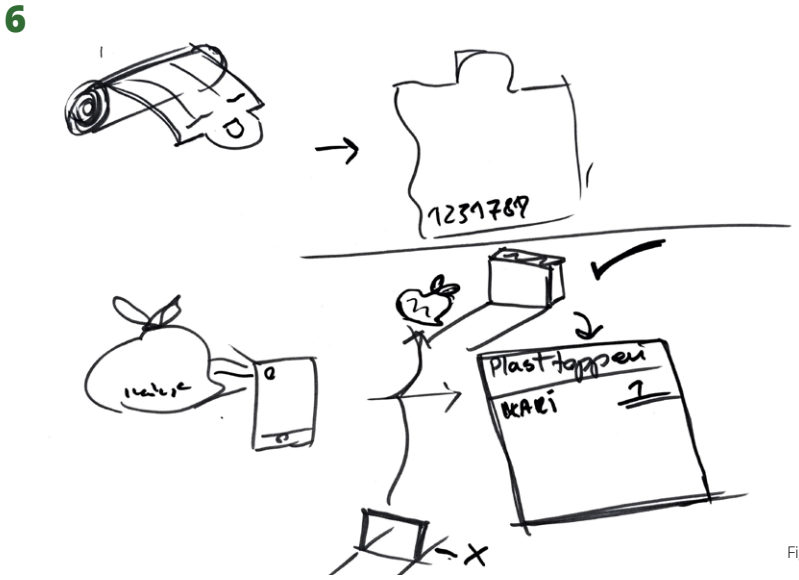
Ideation

5. Scan the waste and receive recycling points that you can share on social media.

6. The roll of blue plastic bags in Oslo can have a unique number that can be scanned in the recycling facility and people can win to encourage people to sort more.

7. Waste collection game inspired by Pokemon Go





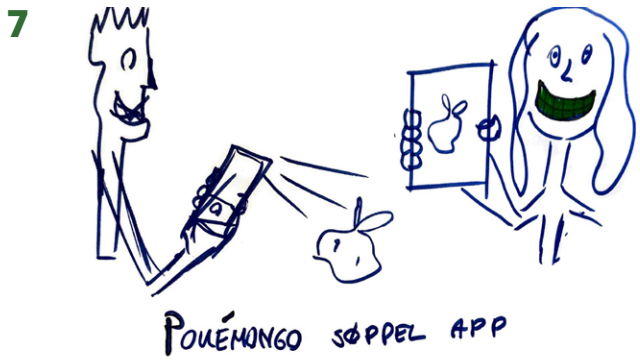


Figure 52: Gamification sketches

Information

Ideation

1. Integrate new waste symbols on waste bins

2. In-store information board where customers can learn about recycling and products

3. Re-naming general waste bin to burn bin so people understands where the waste ends up

4. A waste-weight that helps user not to overfill their waste bins.



Figure 53: Information sketches

Simplify life

Ideation

1. Mini sorting bins that you put over the old once and the empty automatically when you put in a new one. Easy to have on counter when cooking.

2. Stacked waste bins to create more space and easier handling

3. A dish brush that makes it easy to clean food waste from the packaging.

4. A container for glass and metal that makes it easy to empty in the collective bins.



Figure 54: Simplify life sketches

Effective Use of Space

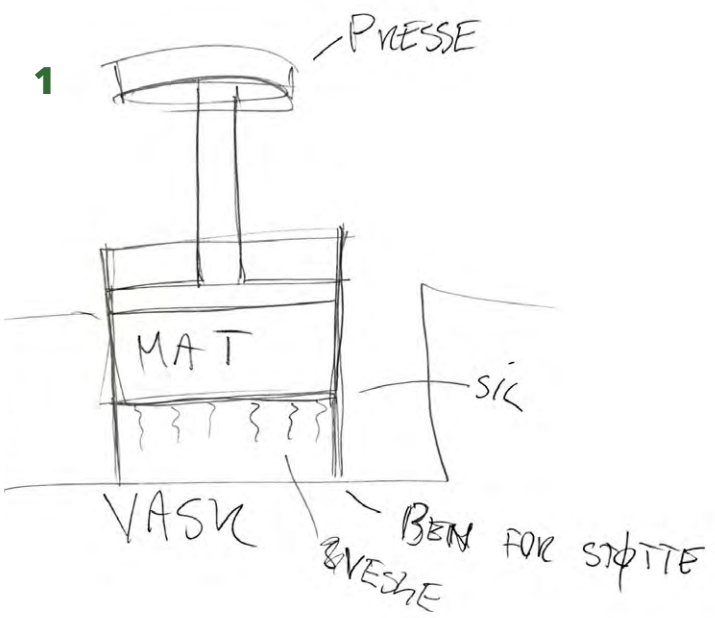
Ideation

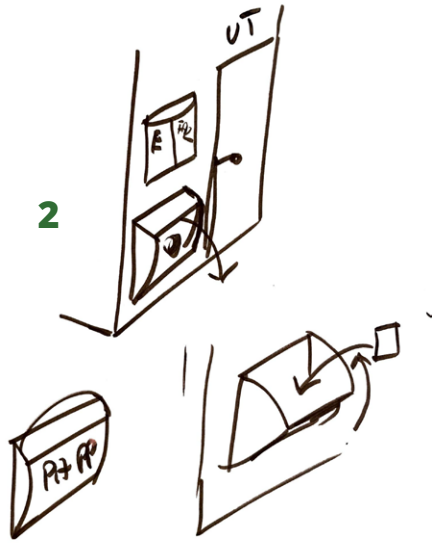
1. A bin for food where you can squeeze the liquid to reduce smell and space of the waste.

2. Paper compressor mounted on the wall

3. Plastic compressor with foot pedal

4. An automatic sorting machine that sort the waste for you into separate bags.





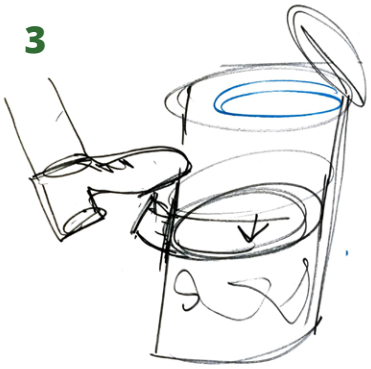




Figure 55: Effective use of space sketches

Future

Ideation

1. Plastic shredder that makes pieces that you can 3D-print into products

2. Toilet as the waste bin. Throw waste in the toilet and flush it out.

3. Waste pick-up service by drones

4. Recycling robots that did collected and sorted the waste from the collection bins

5. Trained rats that sorted waste underground and sewage system

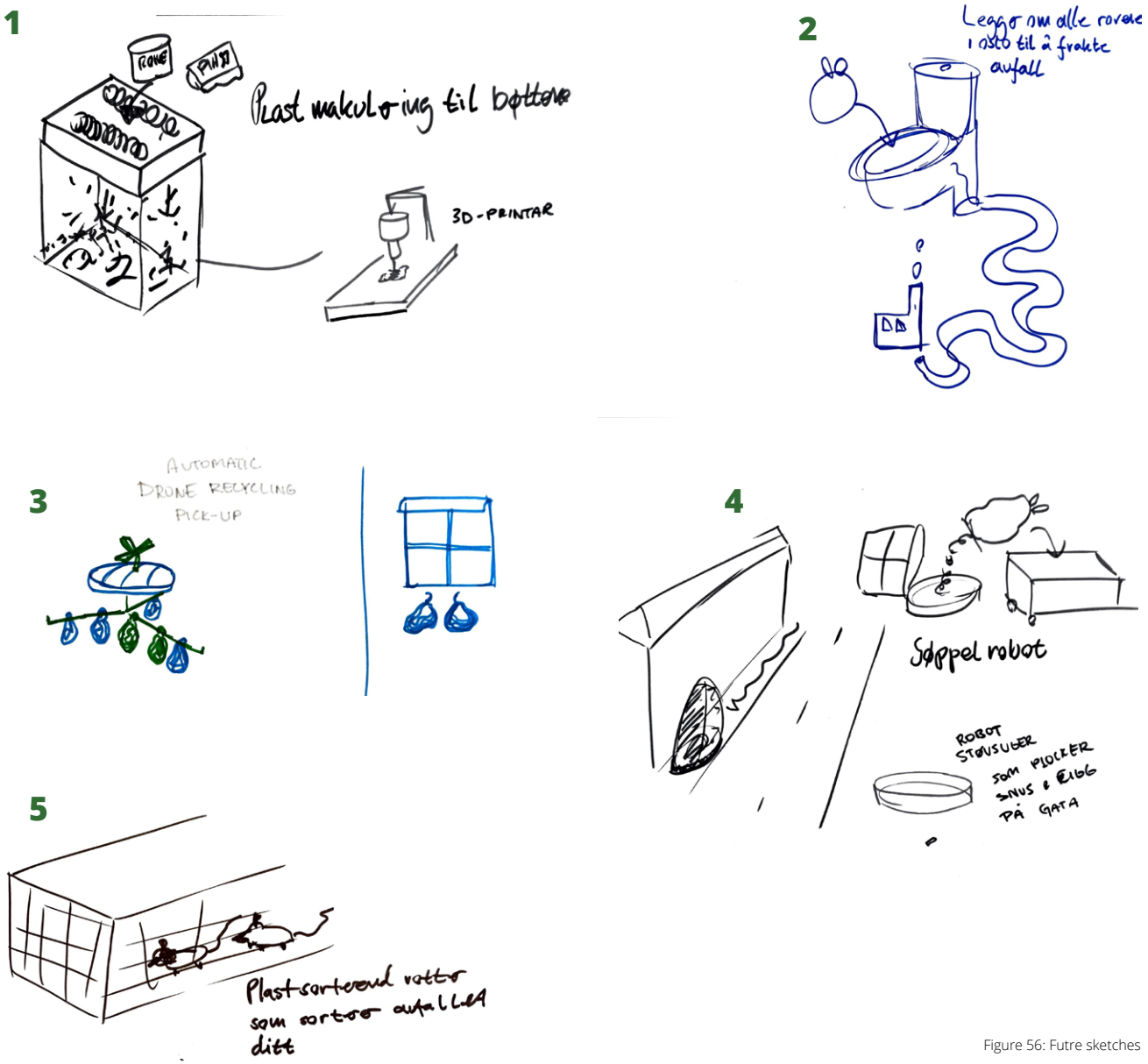


Figure 56: Futre sketches

Idea Selection

Evaluating ideas are important. We therefore invited a family member to be part of the evaluation process (Lerdahl, 2017, p.146-147). Everyone had 10 votes in total and marked the ideas they voted for. After a short discussion we summarized our thoughts and votes.

Based on our wide range of ideas, we selected, combined, and developed the best ones into four different concepts. Each concept is built from a design question that the concept was intended to solve.



Figure 57: Selection Process (Photo: Regina von Koch)



Figure 58: Discussing Ideas (Photo: Regina von Koch)

Concepts



Concept 1

Trash Me if You Can

How to use gamification to get people to pick up waste?



Concept 2

Gamification

How can your personal waste become a game?



Concept 3

Improved Bins

How to develop solutions that use space effectively?



Concept 4

Information Service

How to inform the customer about the waste system, solutions and the importance?



Concept Development

Figure 59: Developing Ideas (Photo: Egle Juodziukynaite)

Trash Me if You Can

Concept 1

Problem Statement

How to use gamification to get people to pick up waste?

Purpose

To bring back waste to the recycling loop from nature.

Concept Description

It builds on the idea of gamification and takes inspiration from the popular mobile game “Pokémon Go”. By using advanced mobile technology, an app can be created where users can locate, collect, and recycle lost waste. Incentives and reward systems can be built into the concept through external partners by creating a community for users through competitions.

Reflection

The concept received a lot of positive feedback from peers. We suggested the potential impact of the concept, especially given that youth and kids are in the primary target market and the focus of environment for younger generations in the future.

The concept had many downsides, how could the service know what people did with the collected waste? There is a change for people to cheat the system, and it was difficult to see the technical aspects working out. Additionally, the concept only partially reflected the key insights as it does not directly improve recycling.

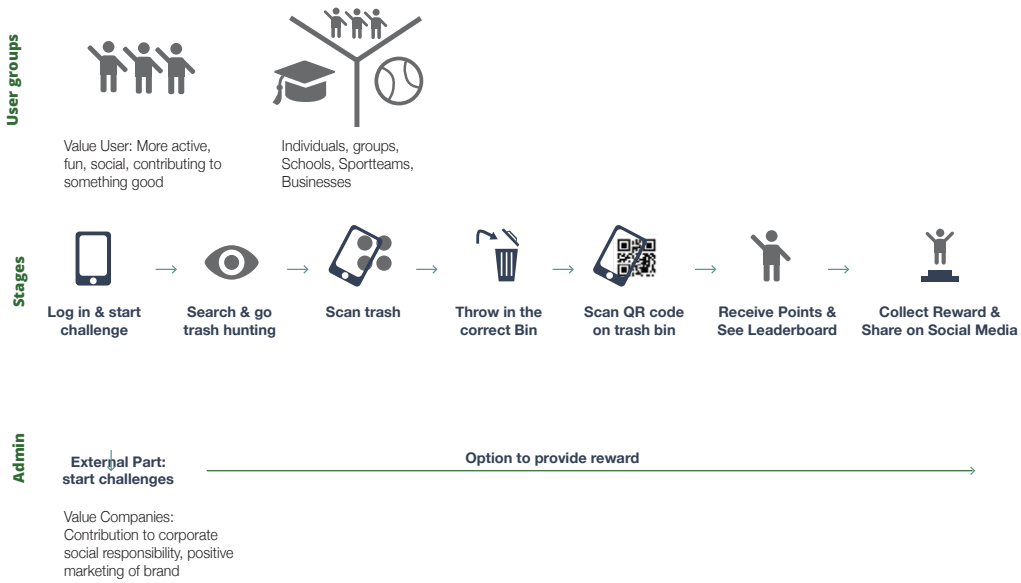


Figure 60: Trash me if you can

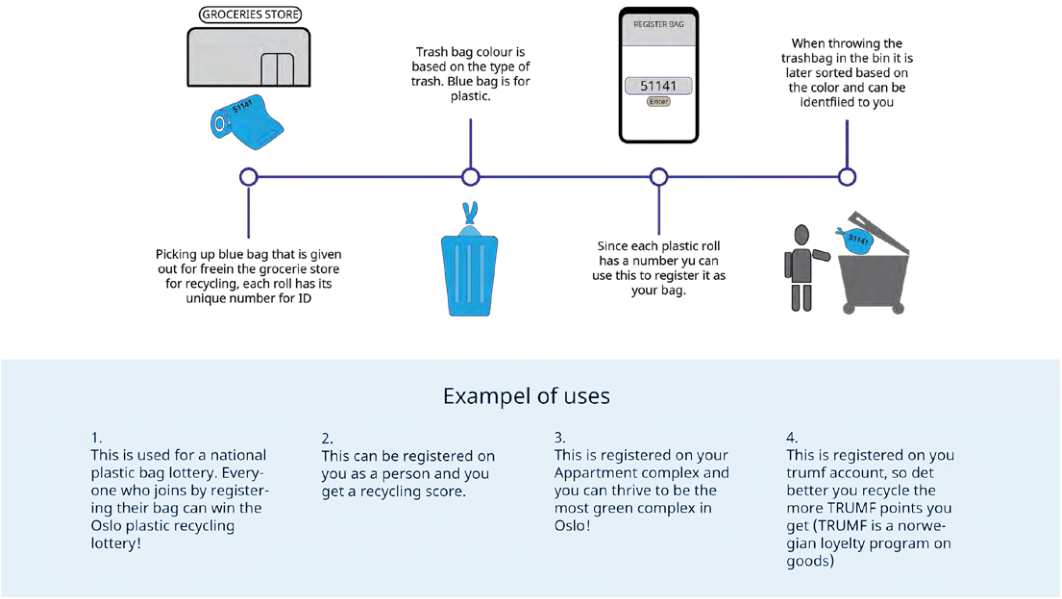


Figure 61: Gamification concept

Gamification

Concept 2

Problem Statement

How can your personal waste become a game?

Purpose

To make more people recycle plastic and make recycling a social activity.

Concept Description

It focuses on tracing waste bags, where the users can participate in games and enter price contests with their traceable bag. By registering the bags, the user can connect the trash bag number to themself and their living complex for a possibility to win money.

Reflection

It may be more time consuming – and not getting people to recycle more. Might not be relevant for IKEA and will not contribute to solve today's issue. When reflecting on this concept, we realized that we thought it was fun, however, it did not match our insights and research, and was not developed further.

Improved Bins

Concept 3

Problem Statement

How to develop solutions that use space effectively?

Purpose

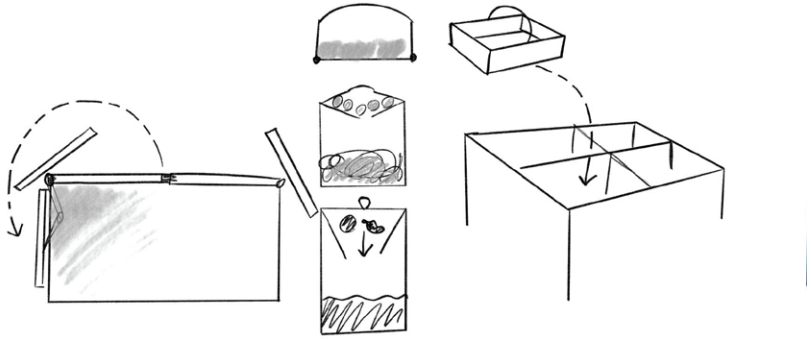
A continuing problem that came up was that recycling takes up too much space. The purpose is to develop solutions that are space effective and making people's everyday life easier.

Concept Description

This concept is looking at developing smart solutions that targets users recycling needs. They include compressing for paper/ plastic and compact systems.

Reflection

Due to the complexity of the recycling system, improved bins may not be as effective as one would think. This is because it is very difficult to create a product that fits with all the different systems. This may lead to high misuse and therefore decrease the rate of recycling.



Smart Lids

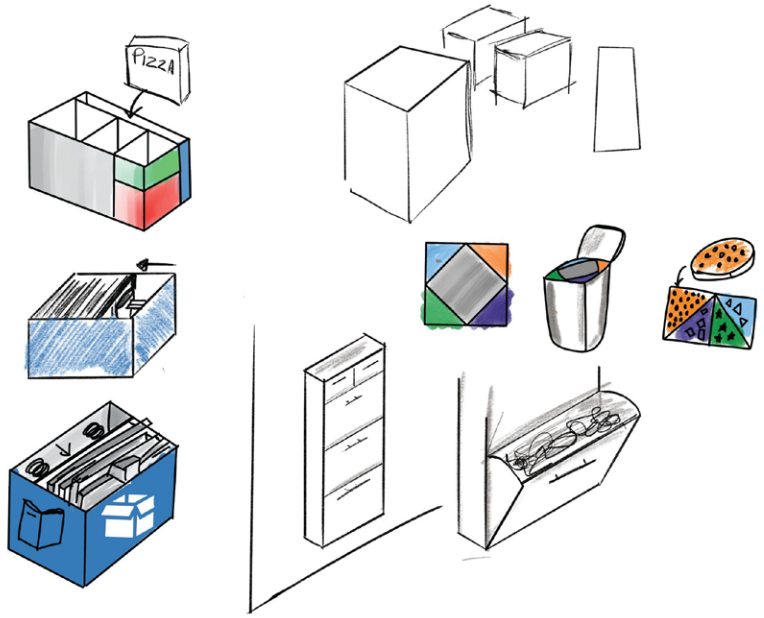


Figure 62: Improved bins sketches

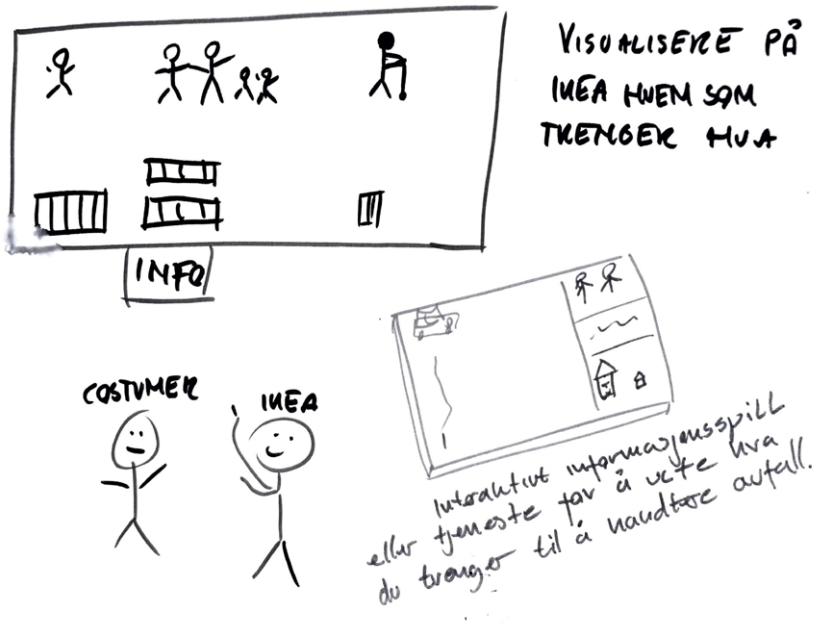


Figure 63: Information sketches

Information Service

Concept 4

Problem Statement

How to inform the customer about the waste system, solutions, and the importance?

Purpose

People have different recycling needs. During the research phase it was discovered that it is chaotic in people homes and the recycling percentage goes down because their system is not optimal. It was also discovered that municipalities have different systems, and different waste types should be sorted depending on where you live. What if there was a service, people could use to discover their recycling needs?

Concept Description

A service to provide home sorting solutions based on discover the customers recycling needs. The costumer will be asked some questions, depending on the answers, recycling solutions will be provided targeting their needs.

Reflection

This might contribute to a big impact on people who buy their new solution. We believe that if people get the optimal system that works for them it will contribute a lot to recycling.

Concept Selection

How

To select the concept which most precisely solves the user's needs, we applied a set of design criteria. It started with the development of these criteria using the method Formulate Criteria's (Lerdahl, 2017, p.180-181). The criteria are based on findings from research, learning outcomes set by the university, and design theory (Brown, 2010, p.18-19).

> *"Innovation has been defined as "is a good idea executed well"*
> - Tim Brown (Brown, 2010, p.111)

Tim Brown defines two characteristics of innovation that we brought with us in the selection process; it must be a good idea and executed well (2010, p.111). We developed a large set of good ideas, and made sure to put forth great emphasis on the execution element in the concept selection phase.

Design Criteria

The Design Criteria was weighted according to their importance and relevance.

Ease of use
How easy is the concept to use?

Feasability
How realistic is it that the concept could be implemented ?

Market Potential
How big is the market for the concept?

Sustainability
How sustainable is the concept?

Increased Recycling Percentage
How does the concept increase the recycling percentage?

Concept Selection

Why

The concepts were evaluated through diagram-based evaluation (Lerdahl, 2017, p.s82-83). Each criterion was weighted based of what is most important for the project. From there we scored each concept on each criterion. In sum, each concept totaled a weighted score which we used as a tool to develop our final concept.

Concept Selection

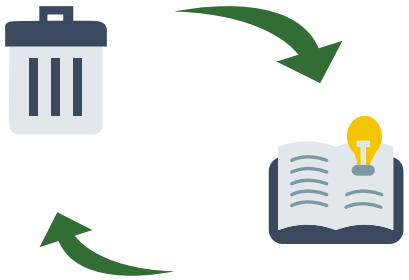
Design Criteria	**Weight**	 **Trash Me if You Can** Score	 **Game-ification** Score	 **Improved Bins** Score	 **Information Service** Score
Ease of use	2	6	2	6	8
Feasability	3	2	4	6	10
Market potential	2	4	4	6	6
Sustainability	3	6	4	6	6
Increased recycling %	3	4	4	4	6
Totals		**22**	**18**	**28**	**36**

Concept Development

The best of two worlds

The Information Service Concept was chosen and developed. By implementing an education element to the purchasing process of recycling solutions, we can impact the user at a time when they are most adapt to retain information. The concept will provide information and services from key stakeholders that is surrounding the home sorting issue; IKEA (the products), local municipalities (the system) and the user (their unique needs).

Through this we will simplify an important part of the complex recycling system and many synergies can be pulled out from this. We find the aspect of identifying interests from recycling stakeholders and collecting them to be at the center of the concept. By creating a "hub" where the user can locate information and services that is relevant, personalized and merged together will help to empower the first line of recycling by providing them with tools (products and knowledge) to make correct sorting decisions.





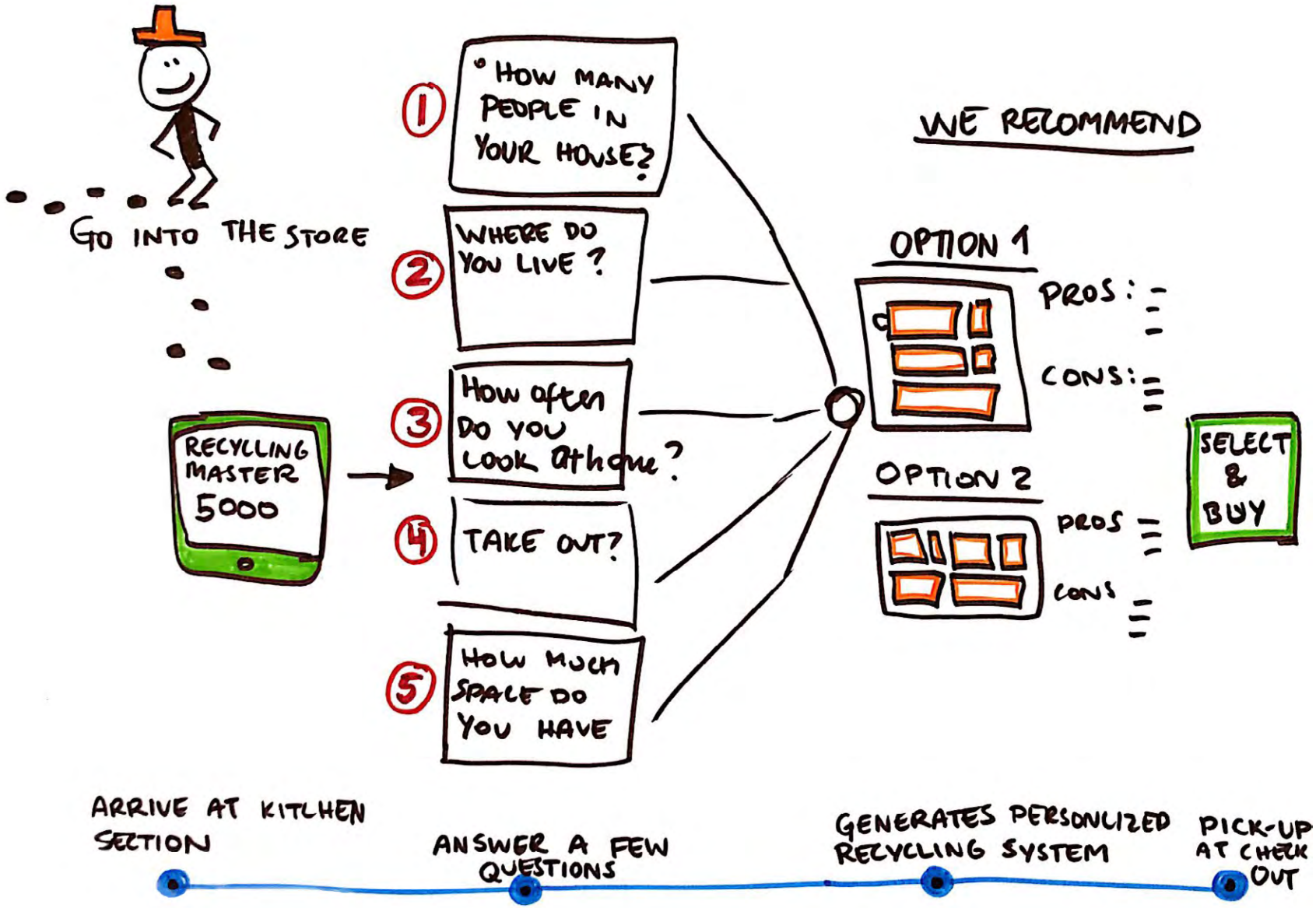


Figure 64: Concept development

Summary of Ideation

Why

Before the ideation phase we had knowledge about the limitations to new products and inventions because of the system. We still wanted to investigate and explore the opportunities that evolved during our research.

It came to our understanding that a lot of products and ideas will make recycling worse because it does not work with the system around it. We could work on changing the system, but with our task from IKEA and with our timeframe available we choose to fix todays problem the best we could.

From an iterative and rigorous ideation and selection phase, we made our way to a service concept that will provide the user with personalized information about local recycling guidelines and home sorting solutions based on their need and situation.

Chapter 4

Prototyping

- Journey Mapping
- Service Blueprint
- Value Creation
- User Testing
- Wireframes



Figure 65: Creating Journey maps

Overview of Prototyping Process

The prototyping process of our selected concept was based on our design criteria and our key insights. Through additional mapping of insights and ideation processes, we prototyped our concepts into services that later turned into a platform.

The core concept of the prototyping process is the user journeys. We made service blueprints and researched existing services to create our first low fidelity paper prototypes. From there, through user testing and iteration, higher-fidelity prototypes were developed - which led the way to our final service design platform prototype. At the start of the prototyping phase a design brief was written, and our final concrete design question was formed to create a stricter work focus.

"How can we improve household recycling by guiding people looking for recycling solutions, to information and offerings based on their household needs, that leads to minimizing general waste?"

Overview of Prototyping Process

Solution goal:
The solution should increase the recycling rate in the home for IKEAs customers. This shall be done through a service where the customer increases their knowledge about recycling and are shown solutions that addresses their recycling needs.

1. User friendly
2. Increase recycling rate of Ikeas existing products through user needs mapping.
3. Make the user more aware of their recycling needs.
4. The service make the customer feel valued when they are buying recycling solutions.

Design goal:
Make a user-friendly service that helps Ikea's customers choose the right recycling solution based on Ikea's existing products.

Target group:
People in search for a new recycling solution. This might be because they are buying a new kitchen, newly moved to a new part of Norway, need help sorting specific type of waste, renovation or in need of a new recycle solution.

Journey Mapping

All Journey Maps

The first step of prototyping the chosen concept was to go back to the insights and start visualizing them in user journeys. The methods used was inspired by the method, “Mapping Journeys” from the This Service Design doing (Hormess, Lawrence, Schneider & Stickdorn, 2011). This is an important step to understand what pain points the customer are facing today to try to solve these.

Home sorting Map

The User Journey map on the next page is based on our insights from interviews and observations done at our home visits in the research phase. It was important to find out what pain point people have in the household to understand their needs for home sorting solutions.



Figure 66: Home Sorting Map (Photo: Øivind Lundestad)

Home sorting Map

Home Sorting

Thoughts & Emotions

Decision	Sorting	Storage	Empty	Collection Bin	Recycling
Frustration over uncertainty of where to throw the waste.	Feels indifferent and ends up throwing it in the general waste bin.	Oh no, the bag has fallen down again.	I dont have time to empty the bins, but they are so full I have to.	It's tiresome to walk to the glass and metal collection bins	How does my contribution actually help?
	It's just one thing - it will be OK				








Pain Points

Decision	Sorting	Storage	Empty	Collection Bin	Recycling
Accesibility and location of bins; they are far away. Difficulty of understanding how to sort the waste.	Bin is full, ends up throwing in wrong bin. Do not have space for different bins so some items are not sorted.	Bin fills up fast and the thrash bags do not fit the bins. Storage area gets dirty.	Time consuming.	Different locations of the collection bins and could be far to walk. Collection bins are full. Glass and metal is dirty to throw away.	

Visits to IKEA

Customer Journey Map

After knowing how the situation was in the household, the next step was to analyze how buying home recycling solution at IKEA looks like today. The customer journey map at IKEA store is based of stories from interviews made at IKEA during the research phase and personal experience.

Primary user

60+ who does not use internet for shopping and needs to go the store.

Reflection

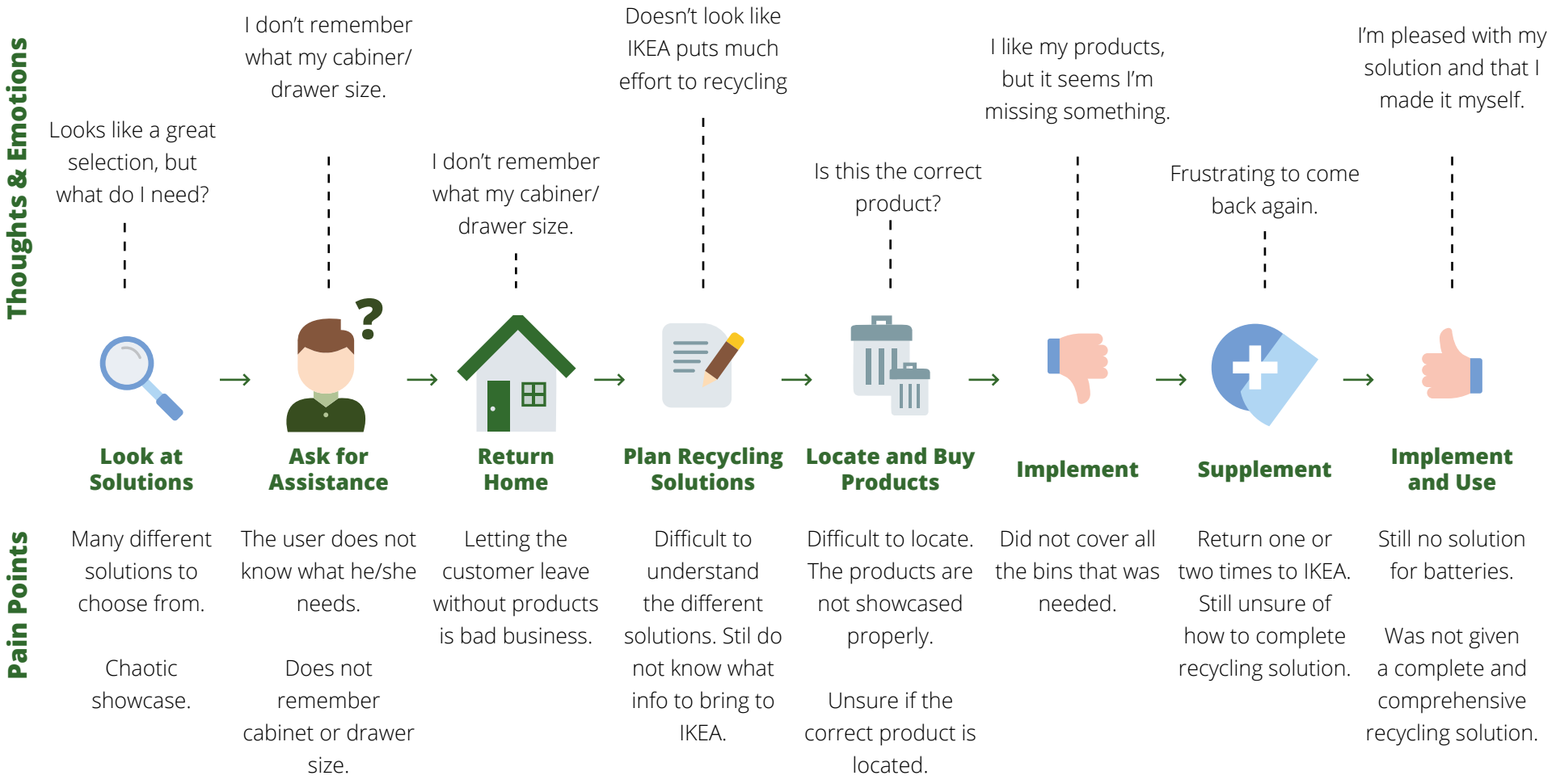
This is a worst-case scenario; however, it has been based on several people have experienced this exact situation.



Figure 67: Mapping Visits to IKEA (Photo: Brage Steen)

Visits to IKEA

Customer Journey Map



Navigating IKEA Website

Customer Journey Map

According to Tim Brown in order to insights at a different level we have to find people, "who live differently, think differently and consume differently" (Brown, 2009, p.44). We decided to create a user journey that is based on a person, that orders everything online with home delivery. We did this to design for two outliers. The map was developed through user testing online via ZOOM, where two people tried to find a recycling solution at IKEA.no.

> *"If you are able to design for these two extremes, then the hybrids like myself, that checks the website and the goes to the store and buy it, is able to use it just fine"*
>
> - Johannes Daae,
> Grønn Punkt

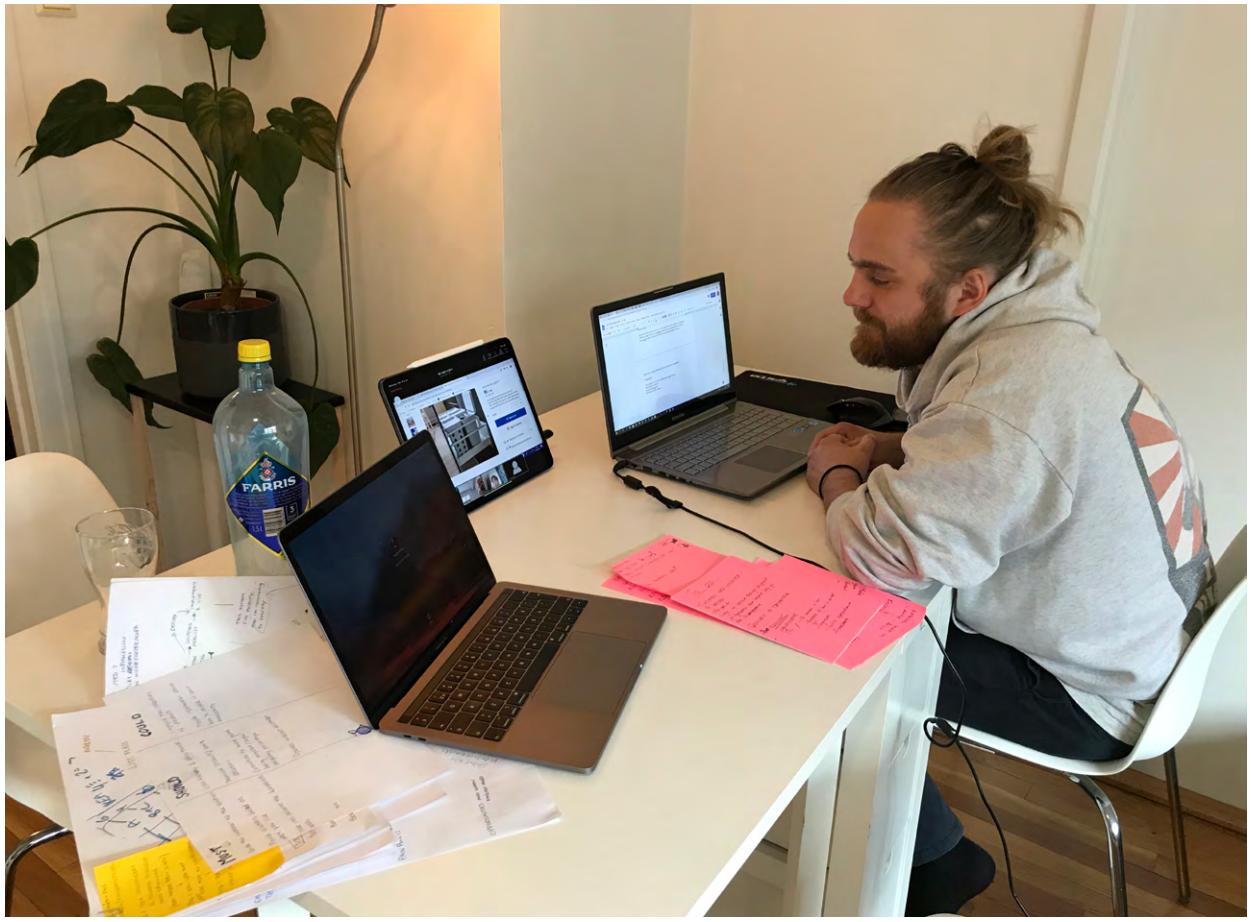


Figure 68: Zoom User Testing (Photo: Regina von Koch)

Navigating IKEA Website

Customer Journey Map

Thoughts & Emotions

"Wow! A lot of options"

"This was advanced. Probably want a package with many bins."

"Why is it so hard to find out what I'm supposed to sort?"

"Need to sort 5 things. That is a problem - it only has 4 buckets"

"I'm so confused. I don't understand how it would fit."

"Is the ones with holes ment to be used for bio waste?"

"This looks familiar, I will go with that one."











Pain Points

Visit IKEA website

Difficult to locate recycling products under categories

Searching for recycling products

No systemized presentation of prodcuts

Browsing local recycling info

Difficult to locate and navigate web for this information.

Measurement and fit

Extremely difficult to find the product measurement.

Select products

Unsure of product fit and style.

Opportunities

Possible to have a recycler helper instead of searching like on kitchen

Only displays kitchen products, not other IKEA sorting products.

Should be diveded into catergories as a standard. "self standing" "cabinet", "drawer" etc.

High threshold to understand - not made for user.

Should be an option for doing this at IKEAs website, like zip code.

Unvertainty about measurement of placement in home

Overview of recycling bins and how to best layout your sorting.

Provide information about product size directly at display. This is equally important as volume.

Building the Service

System Map

We started building the service based on our key finding, pain points in user journeys, and collected data throughout our research. Our first step was to create a system map for our service, where we put together the main elements. The system map was divided into three main parts:

1. Information about User

Since recycling solutions and information is based on individual needs' the service needs to get input from the user. The user needs to provide information about their location and household waste production.

2. Information about Recycling Solutions

To complete the solution, the service need information regarding placement and available space.

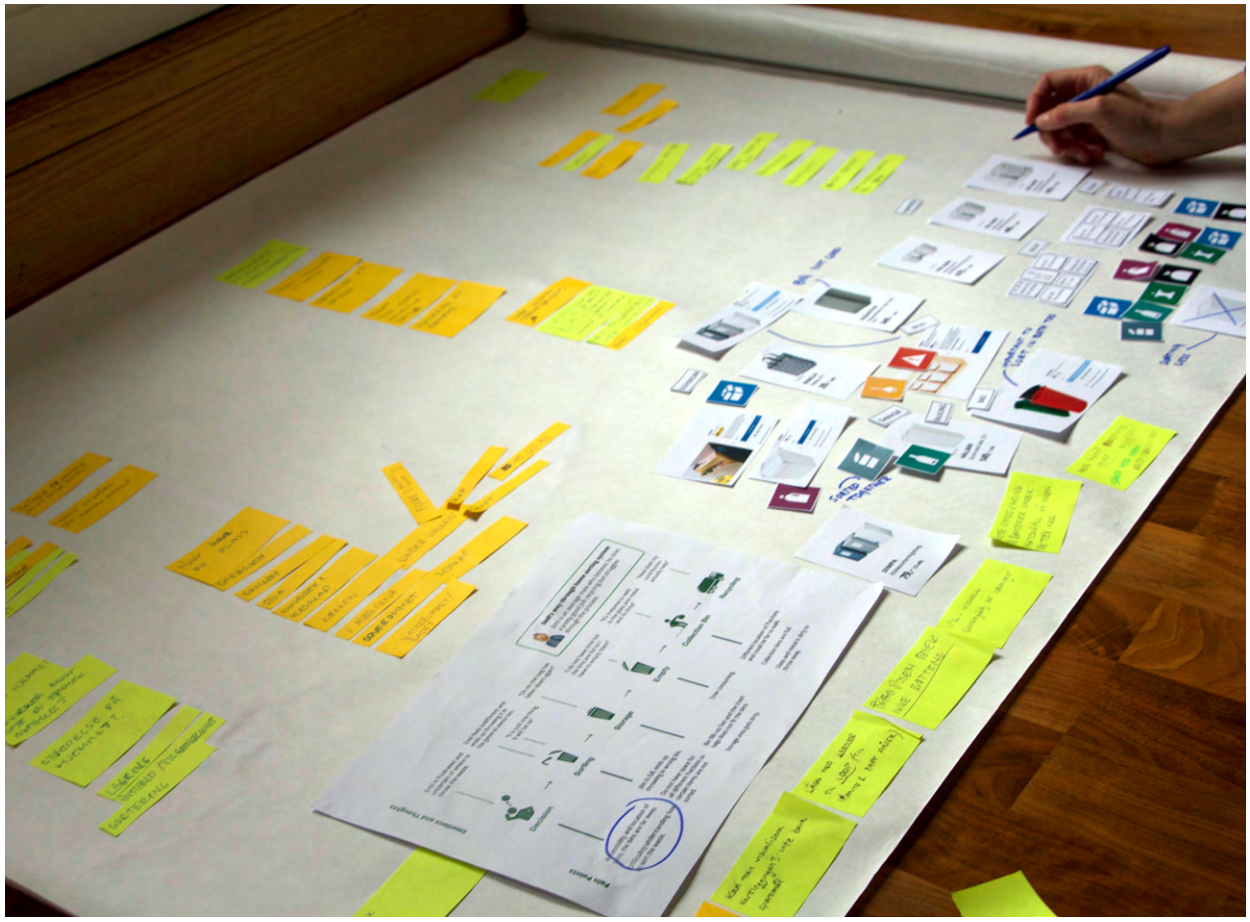


Figure 69: System Map (Photo: Regina von Koch)

Building the Service

System Map

3. Offer a Solution

Show which bins sort what waste. Here it was important to find a solution that fits the user's needs. We printed out different products of Ikea and systemized them like a puzzle.

Our conclusion was that there is not many options of modularity of the solutions under the sink; However, Ikea has a lot of different products where it is possible to sort waste.

Because of the complexity we understood that we needed a service blueprint.



Figure 70: System Map (Photo: Brage Steen)

Service Blueprint

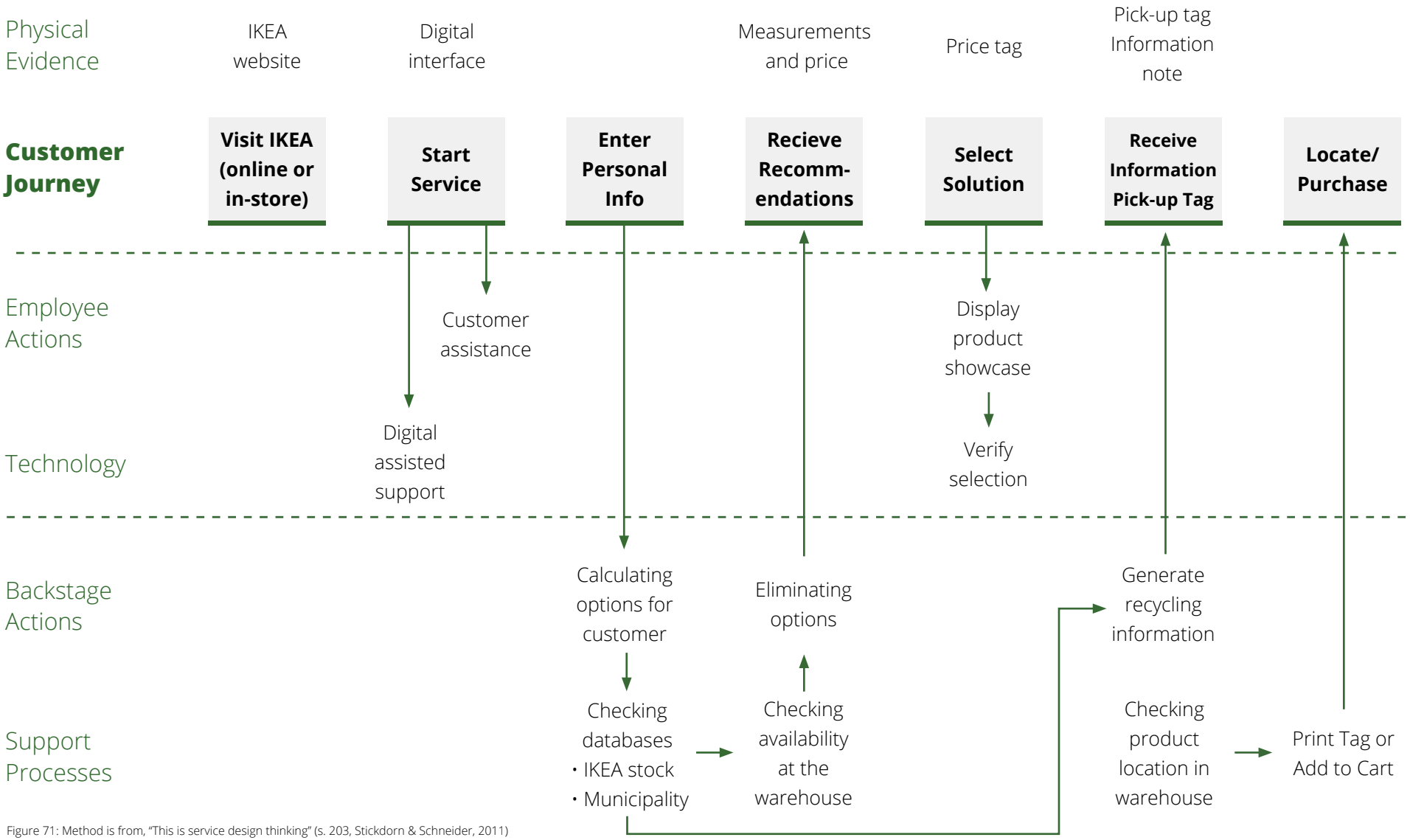


Figure 71: Method is from, "This is service design thinking" (s. 203, Stickdorn & Schneider, 2011)



Figure 72: IKEA Matress Guide (Photo: Screen Dump)

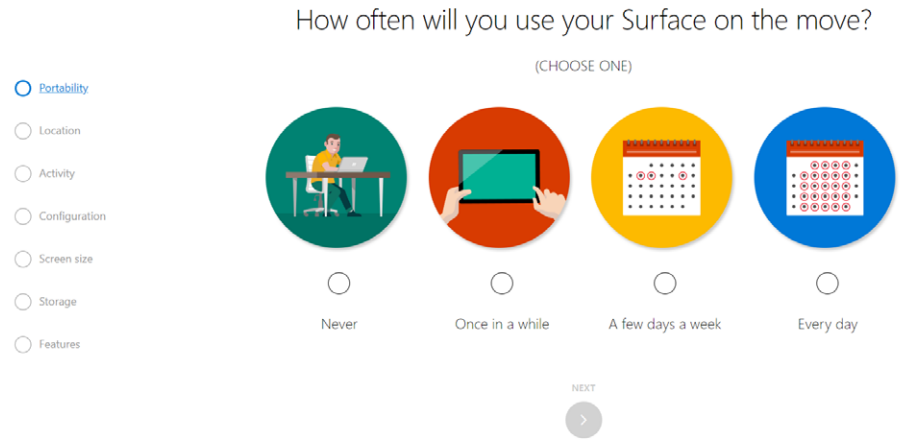


Figure 73: Microsoft Surface (Photo: Screen Dump)

Existing Services

Through analyzing different services online we looked for inspiration on the strengths and weaknesses of how it was built. For inspiration, we found three services that focused on guiding the user to the right solution. We believe this is an increasing trend, because a lot of new products are launched with this service. However, we did not find any existing services providing help for personalized recycling solutions.

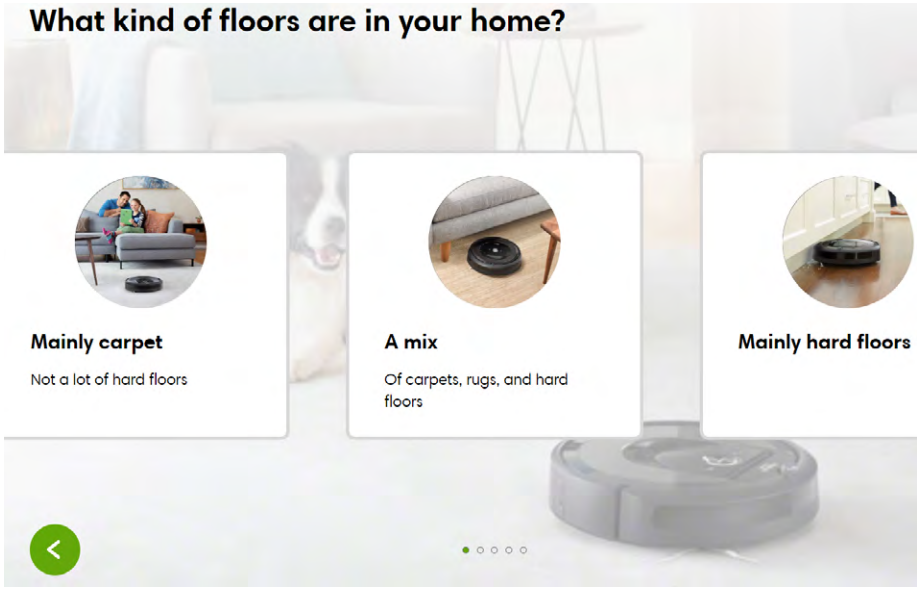


Figure 74: Electric Vacumer (Photo: Screen Dump)

Early Protoypes

Rehearsal Design

We sat down and acted out the service as an interaction between human and service, where humans played the role of the service. Based on methods from Innovating with people (2010, s. 47) and This is service design doing (Stickdorn, M., Lawrence, A., Hormess, M. E., & Schneider, J. 2020)

Paper Prototyping

Based on the collected data from the first prototyping stages, we were able to put together a set of paper prototypes that visualized the service. We have done the exercise, paper prototyping from Service Design Doing, focusing on sketching out the steps of the whole service and understand how we can communicate solution to the user. The exercise started restriction of focusing of designing for a couple that have moved to Oslo.

Reflection

The session made us understand the complexity of the service and we discover more pain points that we need to research further.
How do you know that you have an IKEA kitchen? What do you need to measure to find a working solution? What about pipes?
Ideas that came up, was to provide solutions based on the size of home and different kinds of waste by ranking IKEA Products based on certain criteria, such as small, medium, big housing.

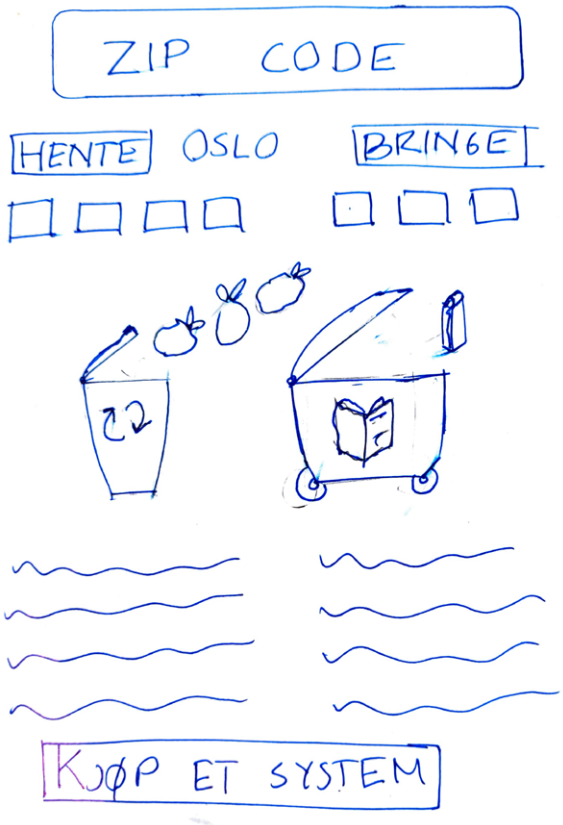


Figure 75: Paper prototyping

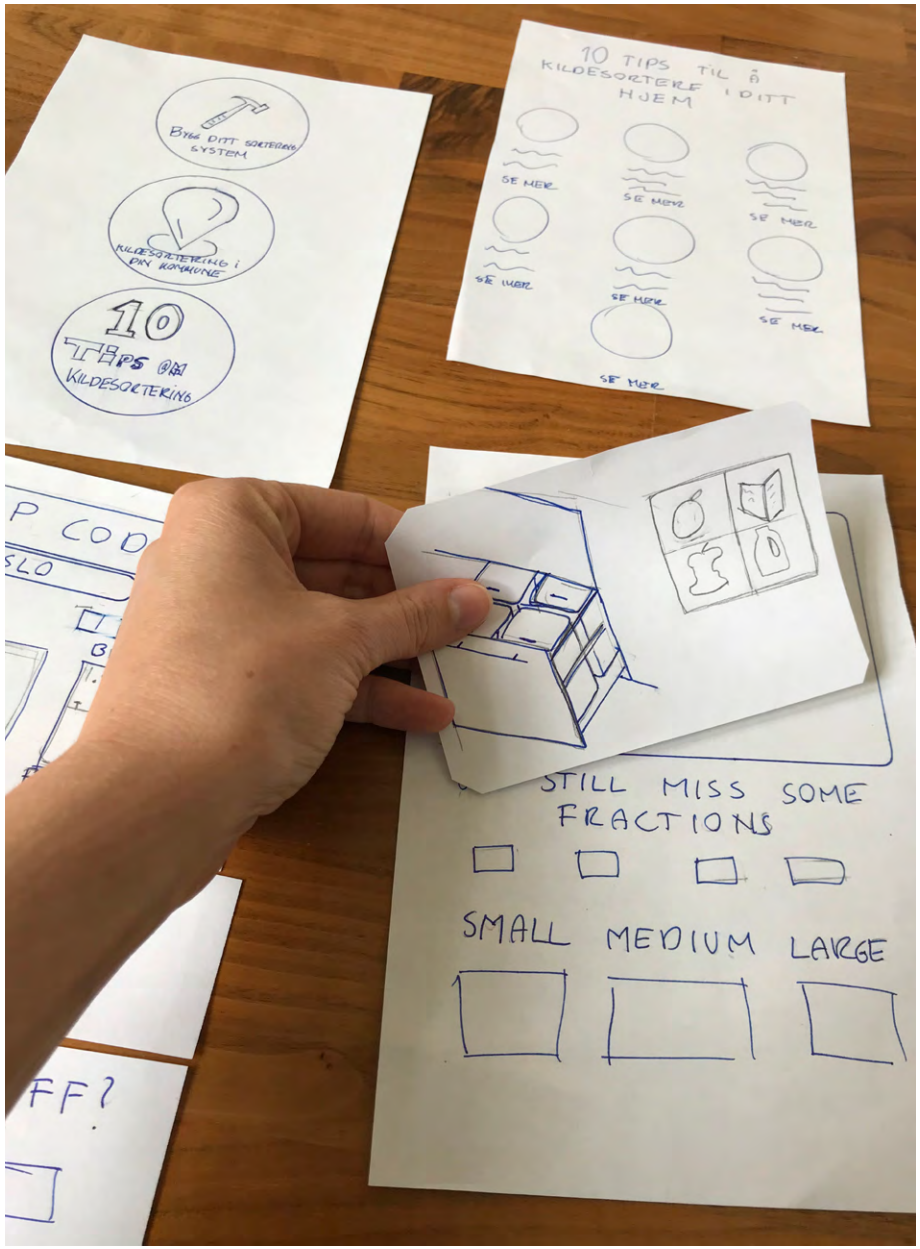


Figure 77: Paper Prototyping (Photo: Regina von Koch)

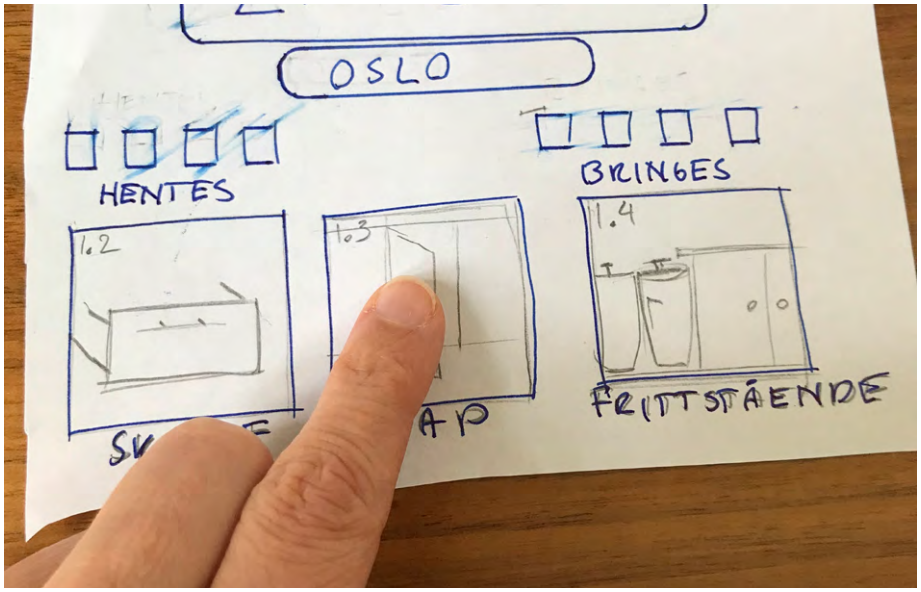


Figure 78: Paper Prototyping (Photo: Regina von Koch)

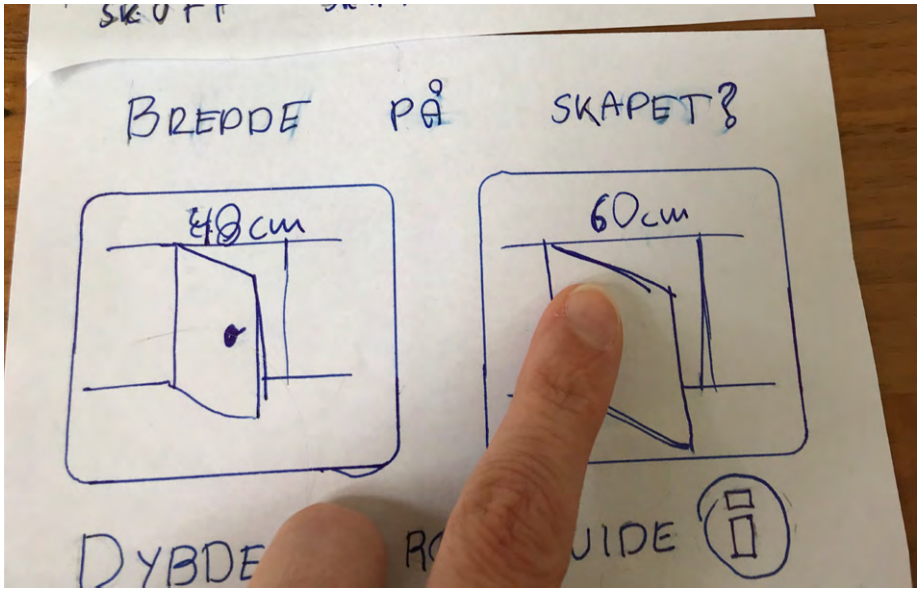


Figure 76: Paper Prototyping (Photo: Regina von Koch)

Co-Creation Session

IKEA Planning Studio

According to Tone Nordbø (2017, s. 97), it is smart to talk to a customer assistant since they are meeting the users daily to find out what kinds of problems the users are experiencing. We visited the IKEA Planning Studio in central Oslo to gain an understanding of what is important when selecting what recycling solutions to choose.

Findings

- The assistant does not ask about user needs - only showcases certain products and solutions.
- Very common to place recycling solution underneath sink where the pipes are a major obstacle for implementation of products
- Estimated 10% of kitchen buyers deliberately asks about recycling.

Reflection

It was important for us to not only focus on what they asked - but also on what they did not ask. By reflecting on this we could discover user needs and pain points. These pain points became a crucial tool for further development of Adobe XD prototypes.



Figure 79: Co-Creation Workshop (Photo: Brage Steen)



Figure 80: Showing Recycling Solutions (Photo: Regina von Koch)

Wireframes

Low fidelity - Paper Prototypes

To test the functions of the prototype, we used an app called Marvel. The prototype can be viewed here:

PRESS HERE

https://marvelapp.com/af6dcdb/screen/68721809

> *"Prototypes should command only as much time, effort, and investment as is necessary to generate useful feedback and drive an idea forward"*
>
> - Tim Brown, Change by Design (2009, p. 91)

Feedback & Road Forward

The prototype was shown to fellow design students and professors, and Anna Kirah service designer at Halogen for feedback.

It is very important what type of language we use; it should be a conversational tone to ease the experience for the user and make it feel "personal".
The most important information should be visualized.

We realized that the basic structure was in place, the next step is to develop the backend and a more visual prototype in Adobe XD.

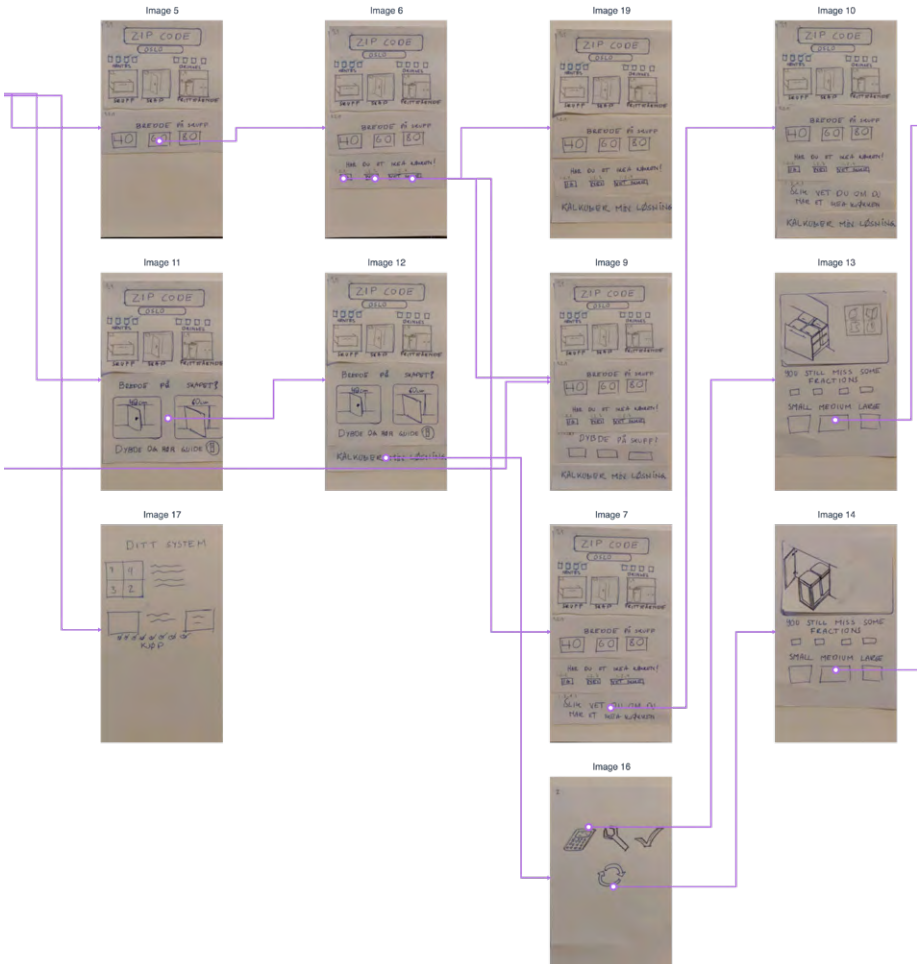
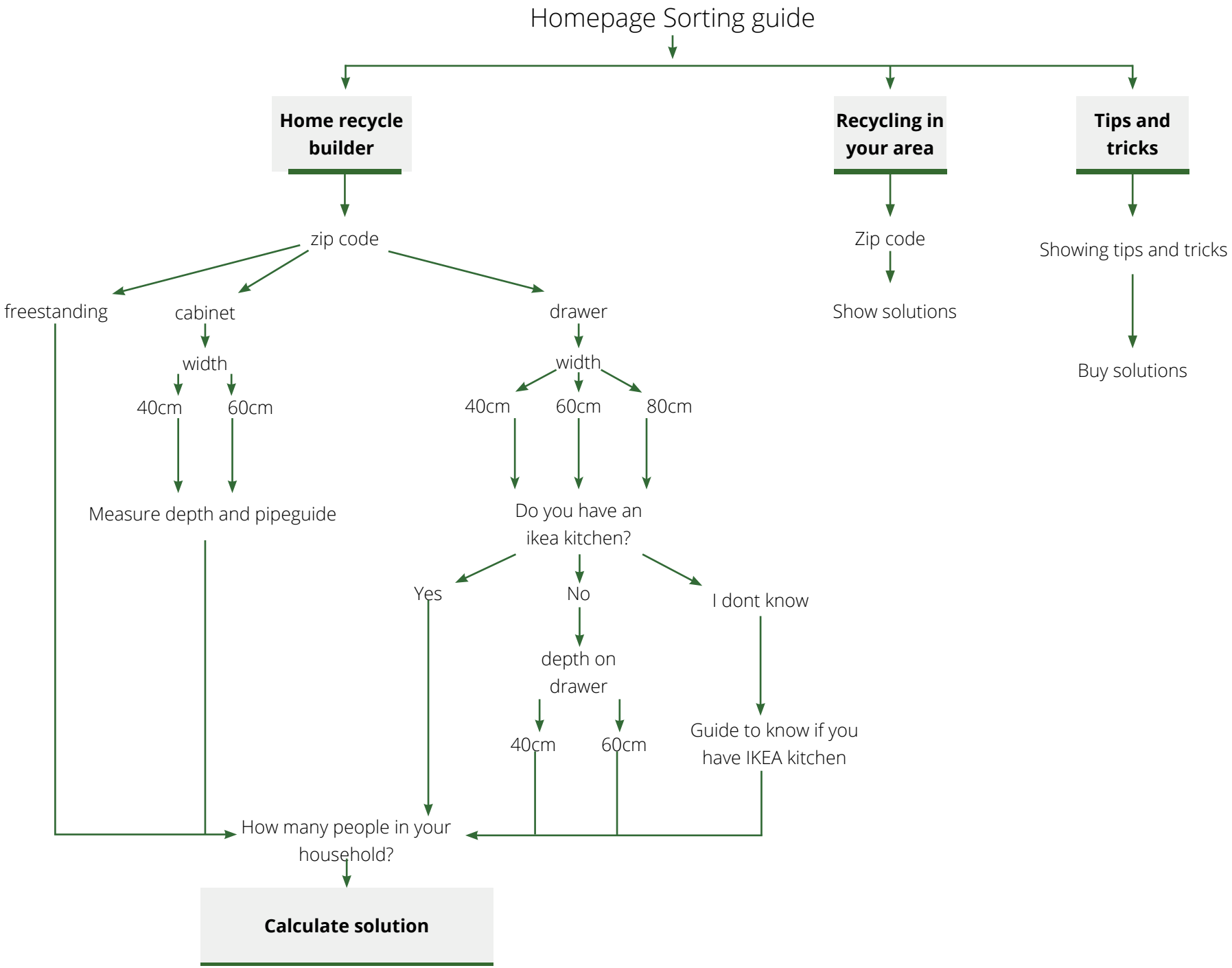


Figure 81: Wireframes showing the connections


Homepage Sorting guide
Home recycle builder
Recycling in your area
Tips and tricks
zip code
Zip code
Showing tips and tricks
freestanding
cabinet
drawer
Show solutions
Buy solutions
width
width
40cm
60cm
40cm
60cm
80cm
Measure depth and pipeguide
Do you have an ikea kitchen?
Yes
No
I dont know
depth on drawer
40cm
60cm
Guide to know if you have IKEA kitchen
How many people in your household?
Calculate solution

Back-end Development

Selection of products

For us to know what products to recommend the user we created a list where we listed the different waste types and what kind of homes they could fit in, Small, medium, and large. The criteria for the different waste types are developed based on our insights. This was based on Rank Data method from Innovating with people (Plumbe et al. 2010, s. 47).

Plastic: Only a case if freestanding. Then it should not be too big because of weight of the bags later in the process.
Food: Needs to be small since a lot food waste is not wanted.
General Waste: Works with the most products.
Paper: Important to have space for it to fit in.
Glass and Metal: Works well with big buckets. Should tolerate some water.
Batteries: Small, should also be glass or metal in case of fire.
Bottles: Needs space and should manage some water spill.

Sizes: We base it a lot on home visits and intuition. Some of the systems take up more space than others and therefore is impossible to fit into smaller apartments.

IKEA PRODUCT	PLASTIC	FOOD	GENERAL WASTE	PAPER	GLASS & METAL	BATTERIES	BOTTLES	SMALL	MEDIUM	LARGE
TRONES	No	No	No	Yes	Yes	No	Yes	Yes	Yes	No
KNALLA	No	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes
DIMPA	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
KARDORNA	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes
KNoDD	No	No	Yes	Yes	Yes	No	Yes	No	Yes	Yes
TROFAST M	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
TROFAST S	No	No	Yes	Yes	Yes	No	No	No	No	Yes
TROFAST SORTERING	No	No	Yes	Yes	Yes	No	Yes	No	No	No
IKEA 365	No	No	No	No	No	Yes	No	Yes	Yes	Yes
IVAR	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes
MJÖSA L	No	No	Yes	Yes	No	No	Yes	Yes	Yes	Yes
MJÖSA S	YES	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
EKOLN	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
FNISS	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	Yes
DRÖNJÖNS	No	No	Yes	Yes	No	No	No	Yes	Yes	Yes
KORKEN	No	No	No	No	No	Yes	No	Yes	Yes	Yes
PLURING	No	No	No	No	No	No	Yes	Yes	Yes	Yes
VARIERA	No	Yes	No	No	No	No	No	Yes	Yes	Yes
FRAKTA	No	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes

Figure 82: Excell list of products.

Abobe XD Early Prototypes

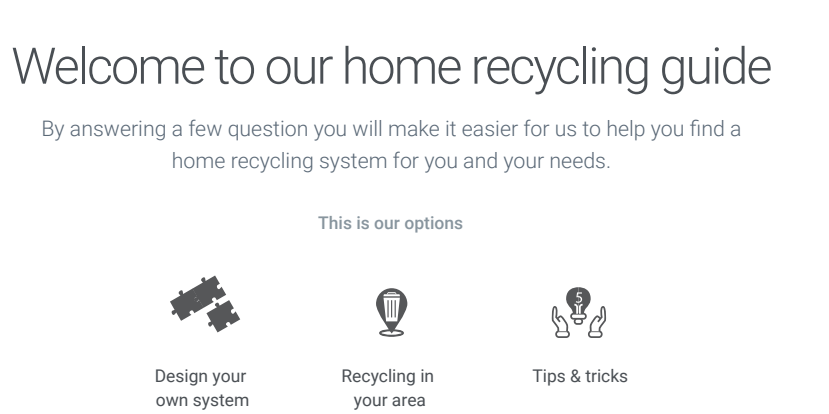


Figure 84: Screen dump service development



Figure 83: Screen dump service development

The focus of the first Adobe XD prototype was focusing on sequence and steps.

Second, we continued the sequence but added visual elements and got feedback from UX designers and users. To see more iteration see Appendix p. 30-31.

User Testing

Throughout the prototyping process we have had a major focus on user testing to effectively discover pain points and possible changes to the design. We have included both expert and novice users in the testing phases, primarily working with OsloMet University students and advisors and our partners from IKEA, Grønt Punkt and Halogen Design, but also selected friends and family.

> *"This made it so much easier to navigate, while choosing home recycling options"*
>
> - 25 year old Student

The testing has been done in several ways; from sending over links, images, and text for testers to work out individually and give feedback on, to interviews, observations, and discussion directly - in-person and online. These different methods of conducting user testing provided us with a broad range of data that was very useful when developing the prototypes. Based on what we learned through user testing, we made changes accordingly.

> *"This was very helpful because I don't know anything about home sorting"*
>
> - 30 year old teacher

Changes

- Using common and personal language, rather than technical and alienating words when interacting with the user
- Using amount of people in household rather than household categories small, medium, and large
- Added functionality Tips & Tricks and Product Guide to provide the user with shorter routes to find their information
- Color choice had to be high contrast to ensure universal design for people who are colorblind
- Remove unnecessary content and tasks on the website - trim down and effective



Figure 85: User Testing (Photo: Regina von Koch)

Value Creation

Service Innovation

By empowering the first-line of recycling with the necessary tools and resources to make correct sorting decisions the service will create added value along the entire value chain.

The environment will benefit greatly from a higher recycling percentage as it reduces the amount of waste lost in the system and increases the amount of re-use which consequently lowers the demand for virgin products.

On the business side the value is apparent; through a more customized product offering, the customer is likely to be more satisfied. Additionally, the service provides an opportunity to up-sell based on the user's need. From a strategic perspective, the IKEA brand and business model can be moved into a greener position and exercise corporate social responsibility.
To further illustrate the value creation of the service, we have connected the value creation to the key findings.



Availability

When people have recycling solutions that are customized to the constraints of the home and the user, the most effective use of space will be achieved. This will directly improve the availability.



Trust

From knowledge comes trust. By building competence through education, the service will build trust to the recycling system and mission.



Education

By incorporating the education aspect to the purchasing process, the service can deliver an important message to the user at a time when they are most adapt to learn and retain the information.

Chapter 5

Implementation

- Presentation of Service
- Universal Design
- Branding
- External Partner feedback
- Business Model Canvas
- Market Strategy

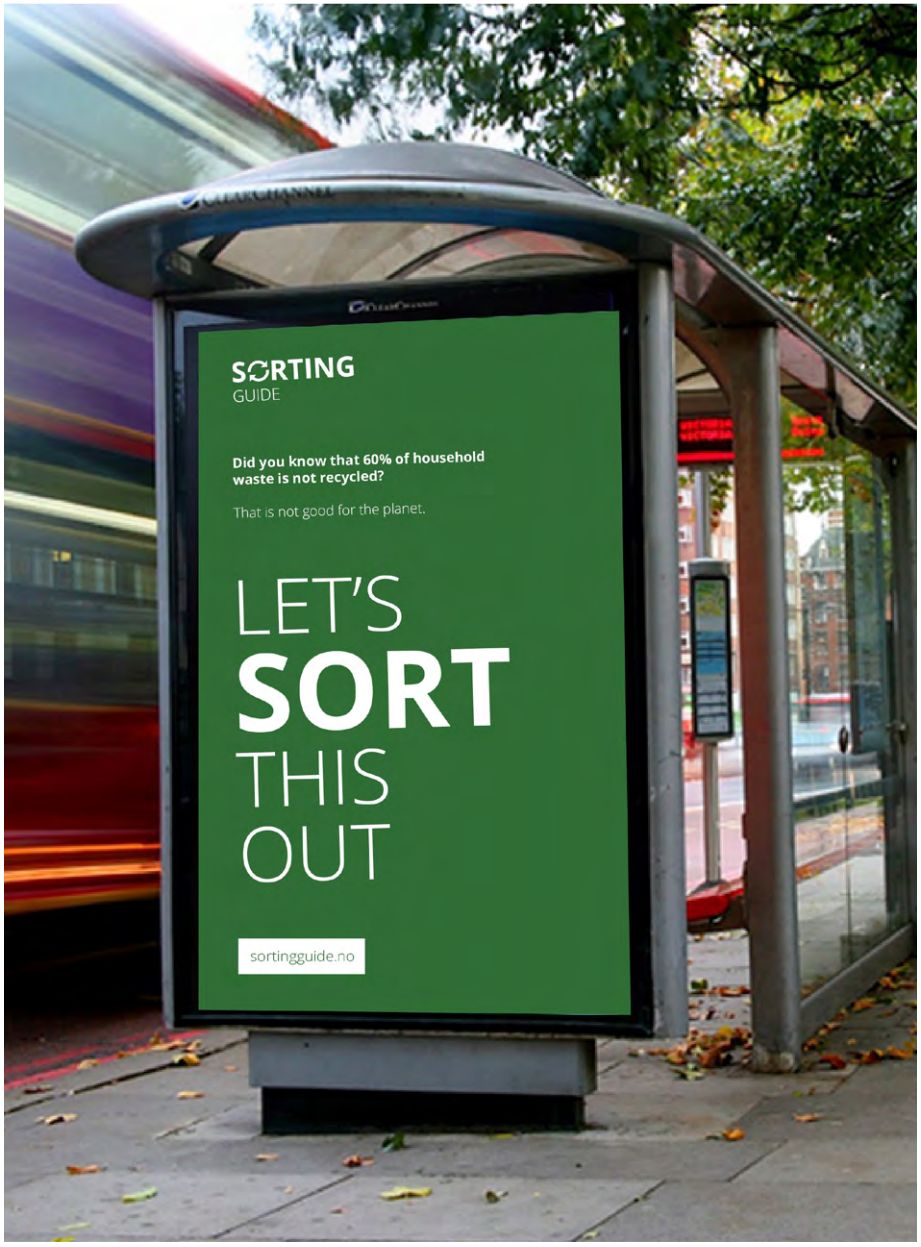


Figure 86: Render showing possible advertisement.

Service Walkthrough

The Sorting Guide service platform consists of four main services that are connected:

- Home Recycling Builder
- Recycling in Your Area
- Product Guide
- Tips and Tricks

These four elements are all sorted under the Sorting Guide platform and provides the user with multiple entry points to the information.

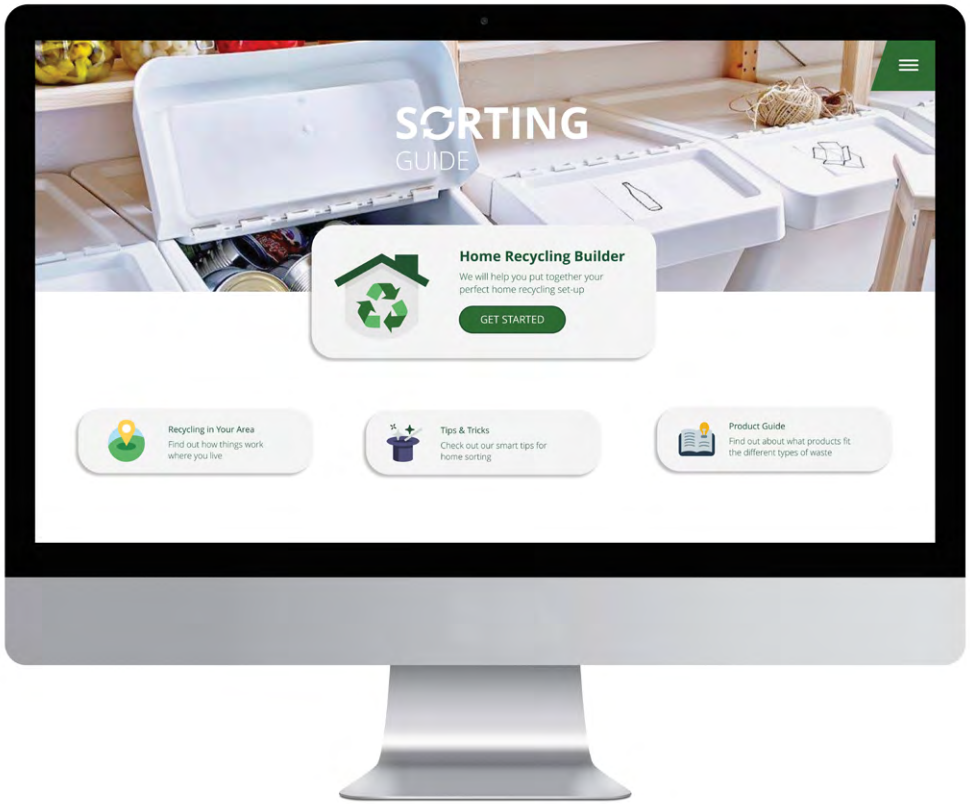


Figure 87: Render of Service Platform

Sorting Guide

Our final developed service is the Sorting Guide platform. The sorting guide is a multi-purpose platform that is both a
1) recycling product online store and
2) educational tool. It serves as a hub where we bring important stakeholders to the recycling issue together to provide the user with the correct tools (home sorting solution and knowledge) to recycle correctly.

The Sorting Guide platform consists of four main services, that all are built on the same data and information but provide the user with different entry points to them depending on their need. These four services are:



Home Recycling Builder

The main element of the Sorting Guide where the user can build its own home recycling system based on their home situation, local municipality requirements and product offering. The user simply enters basic information about themselves and the service provides a recommended home system along with the appropriate instructions for use.



Product Guide

Some users may only need a simple product to fill a specific need. All products in the Sorting Guide are labeled with recycling tags that show which products can sort what type of waste. Through a simple filter-function, the user can simply narrow a massive product offering to few and specific products that fit their need.



Recycling in Your Area

For a quick look-up on the recycling requirements set by local municipalities, we have created an easy-to-use directory where the user can swiftly get information about recycling where he/she lives.



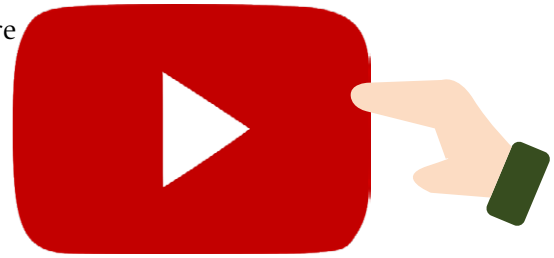
Tips & Tricks

There are many smart solutions and workarounds to make recycling easy and convenient. We have collected these throughout our research and presented them in our Tips & Tricks section.

Video and Adobe XD Prototype

To demonstrate the service, we have developed a video showing how users interact with - The Sorting Guide.

Press here



https://youtu.be/mNr8CHPtbnI

To interact with the prototype

Press here



https://xd.adobe.com/view/60cd8d3c-32ed-435d-7295-8826c24318o7-16fc/?fullscreen&hints=on

Figure 94: Adobe icon. Retrieved from Adobe.com

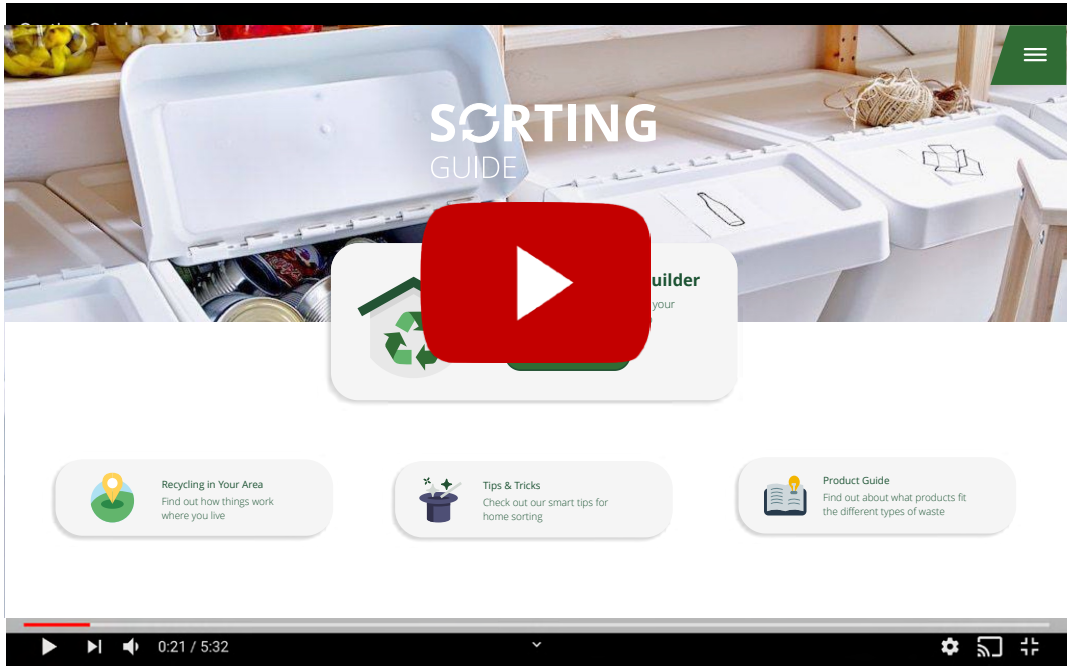


Figure 88: Screendump of service video walkthrough

Universal Design

To increase the user experience, the concept of universal design was applied to the service. Universal design focuses on increasing the accessibility for the user, regardless of their ability (Nordbø, 2017, p.59). In Norway, websites are legally obliged to follow 35 of 61 universal design criteria in the international standard Web Content Accessibility Guidelines 2.0 to ensure this (Digitaliseringsdiektoratet, 2016).

During prototyping we involved UX designer, and web developer, Runar Normark, to make sure that we were fulfilling as many of the universal Design Criteria as possible. These were among the key criteria we focused on:

Easy navigation

Using few colors, large buttons, and by providing the user with positive feedback we were able to simplify navigation.

Readability

By implementing contrasting color, large text size and simple fonts we increased readability.

Input Assistance

By providing the user with smart input suggestions and correction possibilities we help the user to engage with the service without making mistakes.

Keyboard Accessibility

By implementing a function for navigation and using the service through pressing keys on the keyboard we have improved the accessibility

Reflection

We put forth a decisive effort to showcase universal design into the prototyping of our service. However, we do recognize that we still have ways to go before the service becomes a truly universally designed solution. We would like to work more cross-disciplinary with different fields to develop this part of the service.

SWOT Analysis

Strengths

- Provide solutions based of needs
- Simplify the process
- Educate the customer
- Gain trust to the recycling process

Weaknesses

- Limited to system standards. Ex. 60 cm drawer
- Lack of flexibility of existing products
- Human error of input

Opportunities

- Gain trust to IKEA's brand
- Ikea: Not just sell but educate - take a green position in the market
- Upselling

Threaths

- New technology makes today's solution obsolete
- For the service to be in place it needs to work with municipalities - Private and public
- Change in sorting rules - needs to be updated in the service

A further reflection can be viewed in Appendix page 29.



Figure 89: Render of Service on Mobil Device

Is there an app for that?

Sorting Guide on mobile devices

The Sorting Guide platform service will be accessible through mobile devices through an internet browser. We do not want the service to be an application for mobile devices, because the service is something you use when you are in search for help with recycling solutions and not daily. People do not want to download an app for a single time use.

Branding

The Sorting Guide platform implements a green profile to promote values of environment and community. The visuals are simplistic and uncomplicated to effectively communicate the content and message of the platform.

The brand elements are the words Sorting Guide, where the letter" o" in Sorting is replaced with a familiar recycling icon. A typography logo was selected as it is an effective way of communicating its' purpose. We agreed that the implementation of a new icon or standard logo emblem design would deter the brands' professionalism and credibility.

Main logo

SORTING
GUIDE

Black logo

SORTING
GUIDE

Negative logo

SORTING
GUIDE

Figure 90: Development of Logo

Service Advertisement

A key part of the implementation process is the service launch in the marketplace. Through a national marketing campaign named “Let’s sort this out” we hope to catch the attention of the public masses. We liked the slogan as it is a keen play on words that will make people look at it. It functions as a motivational quote to correct environmental issues, while at the same time being a reference on how to fix that problem.

We suggest implementing the advertisement in both digital and print media, and work with local stores to activate the marketing campaign through events or activities that engage the public.



Figure 91: Render of potenial advertisement.

External Partners Feedback

Upon completion of the final prototype, we conducted meetings with IKEA, Grønt Punkt, and Loop, as well as Norway's national sorting information service Sortere.no. These meetings served one main purpose; to validate the service.

Through the validation process we were able to fine-tune the prototype as well as receive feedback that validated our choices down the line. We received a lot of positive feedback and learned a tremendous amount through this final process by putting the pieces together and seeing the simplicity of our service in a complex system.

> *"You are spot on. There is definietly a need for this concept"*
>
> - Kenneth Skauge. 14, may 2020. Sustainability Business Partner at IKEA Group

While our project is tailored to IKEA as our main partner, we discovered that the service is also valuable as its' own entity and that its' positive effects can be multiplied to other companies and industries to serve the recycling mission.

> *"This service tackles the different user groups needs"*
>
> - Olga Korkasheva. LOOP, 14 may. 2020

While validation was our main motive with these meetings, we also had the opportunity to present a sales pitch to these firms that made way for a dialogue moving past the bachelor thesis and into possible realization of the project. They all expressed interest in further discussion and development of the service. In some ways, we considered this to be the ultimate validation.

> *"The home recycling solutions is a main key to recycling, and your services provides the best service to multiple user groups"*
>
> - Sylvelin Aadland, 14, may 2020. Sortere.no

Business Model Canvas

Key Activities



- Educating the user
- Providing personalized home recycling solutions
- Providing information to municipalities
- Building private-public partnerships

What activities do you do?

Key Partners



- IKEA
- National governments
- Local municipalities
- Grønn punkt
- Loop & Sortera.no

Who helps you deliver your activities?

Key Resources



- IT platform
- Knowledge about the complex recycling system
- Relationships with stakeholders

What resources do you own?

Value Proposition

- Improved availability for home recycling solutions
- Builds trust to recycling system
- Increased knowledge for the user
- Improved customer satisfaction
- Increased rate of recycling = better for environment
- IKEA corporate social responsibility

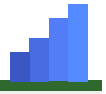
What value do you deliver?

Customer Relationships

- Interactive and online
- In-store assistance
- Brand communication and advertisement

How do you work with the customer?

Channels



- Website (desktop)
- Mobile web
- In-store screen

How do you reach your customers?

Customer Segments



- User
- Municipalities
- Recycling product suppliers
- Environment organizations
- Recycling organizations
- ... and society and the environment itself!

For whom are you creating value?

Cost Structure



- Development Cost
- Implementation Cost
- Operations Cost

Revenue Streams



- Government funding
- Service fees
- Recycling product sales

A thorough financial analysis has been done, see Appendix p. 32.

Business Model Canvas (Osterwalder, Pigneur, 2010)

Market Strategy

Environmental issues are becoming a larger part of public and private businesses, and those who neglect to innovate to meet a greener future is likely to be left behind. Future generations are more focused on societal and environmental matters - like recycling. Understanding this market trend is a key strategy for businesses who will succeed tomorrow.

For IKEA to implement the Sorting Guide platform, they are adapting to this trend in multiple ways:

Corporate Social Responsibility

The platform will directly move the IKEA brand and business model to a greener position in the marketplace, exercising CSR in an area directly connected to IKEAs customers. Though outside the scope of this project, IKEA have an opportunity to create an IKEA GREEN initiative where recycling activities and requirements are implemented in local areas and throughout the value chain.

Public-Private Partnership

IKEA have an opportunity to build and strengthen public-private partnerships, especially with local and national governments, which is very likely to be important in the future. The major issues with recycling are at the system-level and therefore cooperation between private and public entities will be vital.

Ahead of the Curve

An industry disruption is probable to hit the recycling field. The implementation of the Sorting Guide, though it is a solution for today's problem, could begin IKEAs journey to be the disruptor and therefore gain a first-mover advantage, securing them significant market shares and partnerships in the future.

Chapter 6

Discussion & Conclusion

- Future Scenarios
- Reflection
- Conclusion
- Road Forward



Figure 95: Discussion

Future Scenarios

Realization

With many actors that have demonstrated interest in the project, we believe there is a good chance to realize this concept into the market. Based on feedback we may even be able to expand the concept to focus on businesses and how they can build their recycling solutions. This is something we will look into upon the delivery of the project.

User guides

It is an increasing trend to seek information, buy online and have fewer human interactions, as well as having personalized services that cater to the user's specific needs. We believe these elements are valuable in the future of services.

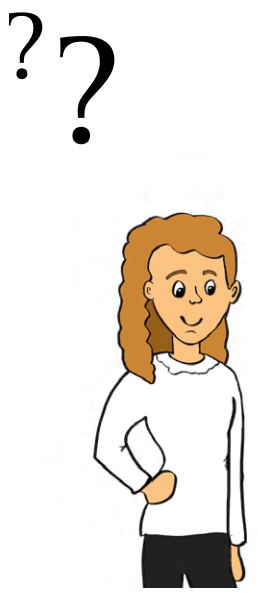
Being able to analyze products and connect them to the right user through a digital service is not only necessary for recycling, but other products and service, paving way for this service to copy over to similar industries.

System change

The recycling systems can best be explained as a system of systems, and it has a high level of complexity. Local differences between municipalities illustrates the complexity in Norway, and when this issue is seen on a global scale, the challenges become obvious. True innovation and disruption in recycling will happen at the system-level. Based on our available time and resources, we decided not to take a system-approach to our solution, but this is something that could be relevant for a potential master project. Through our research we have identified potential improvements that can be implemented to make the recycling system more effective at a global scale.

Reflection

What could we have done differently?



Partner Contact

Throughout the project, we have had great relationships with our partners. Due to the spread of coronavirus, our planned course of action hit a few bumps in the road as the university, our partners and we all had to change to meet a new way of working. This created some challenges for us, but we adapted to the situation and did so by bringing more and other external parties into the project to assist us in the development.

Process Work

In retrospect, we have identified several things we could have improved. We could have had a higher level of user involvement through workshops and we could have been more eager to demonstrate our work to peers. Early on, we were reluctant to showcase our unrefined work. It turned out that the work we believed to be unpolished, was in fact more than enough to receive the feedback we needed. If we showed more unfinished ideas and concepts early, it could have helped us progress more efficiently.

We had a difficult time illustrating our insights and findings along the way. The complexity of the issue and the wide range of collected data made this quite challenging. We spent a lot of time on this issue, hoping to make it easier for the reader to understand the process and logic behind our solution.



Conclusion

Design Question

"How can we improve household recycling by guiding people looking for recycling solutions, to information and offerings based on their household needs, that leads to minimizing general waste?

We believe The Sorting Guide will help people to get a better recycling solution, get more informed about recycling, and thereby increasing recycling rates. By using our service people will get the optimal recycling solution in their home based on their needs. That being said, we have full trust that our Sorting Guide service will be able to improve the rate of recycling, both in Norway and internationally, thanks to the innovations explained in this report.

> *"You are contributing to limiting the largest leakage in the recycling system"*
>
> - Johannes Daae, 14/4-2020
> Responsible EcoDesign at Grønt Punkt

Road Forward

To move this project further, the Sorting Guide needs to be further developed and implemented by IKEA or other stakeholders. We have identified multiple areas where the Sorting Guide itself needs to be developed to fit into existing challenges and structures for it to function properly. These include adaptation to IKEA's kitchen purchasing process, integration with Loop's database on local recycling guidelines and further analysis of IKEA products along with user testing. These elements have already been discussed with IKEA and Loop.

We have been invited to present the solution for IKEA's development team in Norway, and have received interest from Grønn Punkt to attend and present at one of their conferences. Loop has also expressed an interest in a partnership.

We are very excited to see what the future brings for The Sorting Guide.



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List of Figures

All illustrations, if not specified otherwise, are created by the authors of this document.

Figure 4: UN.org. (2020). *SDG 11*. Retrieved from https://www.un.org/sustainabledevelopment/news/communications-material/

Figure 5: UN.org. (2020). *SDG 8*. Retrieved from https://www.un.org/sustainabledevelopment/news/communications-material/

Figure 3: UN.org. (2020). *SDGs*. Retrieved from https://www.un.org/sustainabledevelopment/news/communications-material/

Figure 9: *The four core atívities of Service Design*. Illustration inspired by This is Service Design Doing (Hormess, Lawrence, Schneider & Stickdorn, 2011, p. 185)

Figure 30: Anon. (2020). *Design probe 1*. Retrieved from Anon. 2020.

Figure 32: Anon. (2020). *Design probe 3*. Retrieved from Anon. 2020.

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Figure 36: Hildegunn Iversen. (2020). *User Testing of the new Recycling Icons*

Figure 37: LOOP. (2020). *New Recycling Icons*. Retrieved from Mail

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Figure 66: Lundestad. (2020). *Home Sorting Map*. Retrieved from SD card

Figure 71: *Method is from, "This is service design thinking"* (s. 203, Stickdorn & Schneider, 2011)

Figure 72: Screen Dump. (2020). *IKEA Mattress Guide*.

Figure 73: Screen Dump. (2020). *Microsoft Surface*

Figure 74: Screen Dump. (2020). *Electric Vacuum*